Spectral-Laplacian Prioritized Sweeping: Mathematical Foundations and Theoretical Analysis

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Abstract

Navigating dynamic networks where links arrive or vanish stochastically presents a fundamental dilemma for autonomous decision-making: in congestion-coupled systems, introducing localized shortcuts can trigger Braess's paradox, severely worsening network-wide delays. Model-based reinforcement learning planners using canonical Prioritized Sweeping (PS) suffer from myopic Bellman updates: perceiving an emergent shortcut as zero-cost, they enter it greedily before downstream bottleneck cascades can propagate backward.

To overcome this limitation, we present **Spectral-Laplacian Prioritized Sweeping (SL-PS)**, a principled framework unifying spectral graph theory, Wardrop equilibrium flows, and canonical dynamic programming over general directed networks. Establishing that the regularized out-Laplacian Green operator is algebraically identical to the discounted Successor Representation, SL-PS projects downstream congestion divergence forward via an instantaneous resolvent value warm-start ($V^{(0)} = V^*_{\text{base}} + \beta\Phi_{\text{spec}}$). This achieves deterministic trap immunity at zero sweeping budget ($K = 0$) while strictly preserving the canonical Bellman optimality operator. Backward in-Laplacian Green potentials subsequently schedule Bellman backups to eliminate upstream priority starvation.

Theoretically, we prove: (i) strict γ-contraction and Banach convergence of canonical value iteration; (ii) deterministic trap immunity under an exact local advantage threshold that naturally vanishes in heavy traffic ($\mathcal{O}(c_{\max}/q) \to 0$); (iii) Pareto-optimal policy refinement closing sub-optimality gaps at geometric rate $\mathcal{O}(\gamma^K)$; and (iv) an unbounded asymptotic travel-time advantage over canonical PS scaling as $\Omega(q \cdot \text{diam}(G))$. SL-PS resolves the fundamental trade-off between decision-time safety and long-term optimality in non-stationary congestion games.

1 Introduction

Model-based reinforcement learning (MBRL) and dynamic programming offer compelling sample efficiency by leveraging transition dynamics to plan offline before acting [15]. Among classical methods, *Prioritized Sweeping* (PS) [11, 10] accelerates value convergence by maintaining a priority queue ordered by the magnitude of Bellman residuals $|\mathcal{T}V(s) - V(s)|$. By focusing computation on states experiencing significant value discrepancies and propagating updates upstream to predecessors, PS achieves orders-of-magnitude faster replanning than uniform value iteration.

However, in non-stationary networks where topology evolves dynamically and edge latencies depend endogenously on aggregate traffic flow, canonical Prioritized Sweeping encounters a fatal failure mode. Consider an autonomous vehicle navigating an urban road grid or a robot in a fulfillment warehouse. When a shortcut link dynamically opens, the nominal traversal time across the new link appears negligible. Standard PS immediately observes a large negative residual at the shortcut link, assigns it high priority, and propagates this apparent cost reduction upstream.

Crucially, in multi-user congested environments, selfish background traffic also diverts into the newly opened link, governed by Wardrop User Equilibrium [16]. As demonstrated by Dietrich Braess [4], introducing network capacity in congested settings can trigger *Braess's Paradox*, where traffic swarming the shortcut severely overloads adjacent downstream and upstream corridors, increasing travel latency for everyone.

Because canonical Bellman updates propagate one hop at a time through realized transitions, the planner experiences an information lag: it plans an eager trajectory toward the shortcut before downstream congestion signals can propagate back. Under limited planning budgets, the agent commits to the deceptive shortcut and becomes trapped in the bottleneck.

1.1 Overcoming Myopic Backups via Spectral Graph Theory

Recent work on Graph Topology Augmentation for Prioritized Sweeping (GTA-PS) [6] demonstrated that directed Laplacian Green's potentials can proactively diffuse value residuals across bottlenecked regions. Concurrently, Lu and Miller [8] analyzed conditions under which uniform affine rectangular grids $G_{m,n}$ manifest Braess's paradox in relation to electrical voltage drops under the grid Laplacian ($V_{uv} < 0$) and level gain ($k > 0$). In spectral graph theory, foundational advances by Spielman [14] provide analytical tools for electrical dissipation, Dirichlet energy, and Green's functions on general networks.

Synthesizing these insights, a fundamental question emerges:

> *Can we formulate a mathematically principled dynamic programming framework that couples canonical Bellman planning with global Laplacian spectral resolvents, enabling an agent to anticipate and avoid Braess bottlenecks across general, arbitrary network topologies without heuristic value corruption?*

1.2 Contributions

We present **Spectral-Laplacian Prioritized Sweeping (SL-PS)**. Our core contributions include:

1. **General Problem Formulation**: We formalize the *Congestion-Coupled Dynamic Graph MDP* (CC-DG-MDP) on arbitrary directed networks $G = (V, E_0)$ with unconstrained topological edge-transition actions $\mathcal{A}(u) = \mathcal{N}_{\text{out}}(u)$, establishing the mathematical interface between discrete stochastic agent transitions and continuous Wardrop traffic re-equilibration.

2. **Spectral Resolvent Value Warm-Start & Safety-Optimization Duality**: Rather than corrupting the Bellman optimality operator with ad-hoc heuristic penalties, we establish that with spectral regularization $\eta = \frac{1-\gamma}{\gamma}$, the directed out-Laplacian Green operator $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1}$ is algebraically identical to the discounted Successor Representation resolvent operator $\gamma(I - \gamma P)^{-1}$. Upon chord emergence, SL-PS executes an instantaneous Tier-1 resolvent value warm-start $V^{(0)} = V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}$ driven by the unsigned divergence $\xi_{\text{div}} = |B\Delta\vec{\ell}| \geq 0$, guaranteeing deterministic trap immunity at $K = 0$. Because ξ_{div} is sign-blind on congestion-relieved bypass corridors ($\Delta f_e < 0$), SL-PS subsequently executes $K \geq 1$ Tier-2 canonical Bellman backups $\mathcal{T}$ with signed physical transition costs to unlock the relieved optimal bypass.

3. **Theoretical Guarantees on General Graphs**: We prove that the canonical Bellman operator $\mathcal{T}$ is unconditionally a strict γ-contraction in L_∞ on arbitrary finite graphs (cyclic or acyclic), preserving the true physical interpretation of V^* as the exact optimal expected travel time. We establish an a priori, non-circular Braess Trap Immunity Theorem for general topologies. Furthermore, we derive the Lipschitz Stability Theorem, establishing explicit robustness tolerance margins under inexact Wardrop equilibrium estimation, and prove the Asymptotic Dominance Theorem under multi-stage corridor renewal dynamics, showing that the cumulative travel time advantage of SL-PS over Standard-PS diverges to $+\infty$ at rate $\Omega(q \cdot \text{diam}(G))$.

4. **Bi-Directional Priority Scheduling (Sweeping Efficiency & Non-Starvation)**: To solve the priority starvation of upstream diverters under finite backup budgets K, we integrate bi-directional Green's potentials ($f_{\text{out}} = \mathcal{G}_{\text{out}}\xi_{\text{div}}$, $f_{\text{in}} = \mathcal{G}^T_{\text{in}}\xi_{\text{div}}$) modulated by the Second Largest Eigenvalue Modulus (SLEM) $\sigma(P) = |\lambda_2(P)|$. This guarantees instantaneous spatial reachability ($\text{Radius}_{\text{SL-PS}}(0) = \text{diam}(G)$), ensuring upstream junctions receive timely updates before agent arrival.

5. **Comprehensive Mathematical Verification**: We provide complete, mathematically rigorous proofs for all spectral resolvent bounds, M-matrix Neumann expansions, canonical operator contractions, non-circular immunity thresholds, and asymptotic divergence under dynamic branching, establishing an unassailable mathematical foundation for non-stationary planning in congested networks.

2 Related Work

Prioritized Sweeping in Model-Based RL. Prioritized Sweeping was introduced by Peng and Williams [11] and Moore and Atkeson [10] to accelerate tabular dynamic programming. Subsequent extensions have incorporated prioritized experience replay in deep Q-networks [13]. However, these algorithms assume stationary or exogenous reward signals, leaving them ill-equipped for endogenous congestion where actions induce systemic equilibrium shifts.

Spectral Graph Theory and Laplacians. The study of graph Laplacians $L = D - A$, pseudoinverses L^+, effective resistance R_{eff}, and Green's functions is foundational in spectral graph theory [14]. In RL representation learning, proto-value functions (PVFs) [9] employ undirected Laplacian eigenvectors as smooth value approximation bases. Recently, GTA-PS [6] introduced regularized directed Laplacians to diffuse Bellman residuals in non-stationary reward environments.

Braess's Paradox and Congestion Games. First discovered by Dietrich Braess [4], the paradox has been analyzed through algorithmic game theory, non-atomic congestion games, and the Price of Anarchy [12]. In route-choice modeling, Zhou et al. [18] investigated how day-to-day commuter choices converge to Wardrop equilibrium under a stateless Bush–Mosteller learning automaton. Lu and Miller [8] proved exact analytical bounds on directed rectangular grids $G_{m,n}$,

establishing that zero-latency chords maximize the Braess ratio. Our work bridges this literature into real-time autonomous agent planning: rather than simulating passive day-to-day population convergence, we formalize active, single-agent dynamic programming under non-stationary congestion shifts.

3 Problem Formulation: Congestion-Coupled Dynamic Graph MDP

3.1 General Network and Congestion Architecture

Let $G = (V, E_0)$ denote an arbitrary finite directed network with vertex set V ($|V| = N$) and directed base edge set $E_0 \subset V \times V$ ($|E_0| = M$). Every base edge $e = (u \to w) \in E_0$ carries an affine, load-dependent latency function:

$$\ell_e(f_e) = a_e f_e + b_e, \quad a_e > 0,\ b_e \geq 0, \tag{1}$$

where f_e is the aggregate background traffic flow traversing link e. An aggregate background traffic demand $q > 0$ enters the network at origin source $s \in V$ and exits at destination sink $t \in V$, inducing a net injection vector $p = q(e_s - e_t) \in \mathbb{R}^{|V|}$.

Let $B \in \mathbb{R}^{|V| \times |E_0|}$ denote the directed incidence matrix of the network:

$$B_{v,u\to w} = \begin{cases} +1, & \text{if } v = u, \\ -1, & \text{if } v = w, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Let $W = \mathrm{diag}(1/a_e)_{e\in E_0} \in \mathbb{R}^{|E_0|\times|E_0|}$ denote the edge conductance matrix. The weighted graph Laplacian of the active base network G is $L = BWB^T$. Under net traffic injection p on an interior equilibrium support where free-flow offsets satisfy potential balance ($BWb = 0$, e.g., uniform coordinate latency offsets across topological levels), the background electrical potential vector $\phi \in \mathbb{R}^{|V|}$ is given by $\phi = L^+ p = qL^+(e_s - e_t)$, where L^+ denotes the Moore-Penrose pseudoinverse of L (for general heterogeneous offsets b_e, the potential field evaluates to $\phi = L^+(p + BWb)$).

3.2 The General CC-DG-MDP Environment

We formalize autonomous decision-making as a **Congestion-Coupled Dynamic Graph MDP** $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{C}, \gamma)$:

- **State Space**: Physical vertex $s_t = u \in V$. Destination sink $t_{\text{sink}} \in V$ is a cost-free absorbing state satisfying $c(t_{\text{sink}}, a) = 0$ and $\mathcal{P}(t_{\text{sink}} \mid t_{\text{sink}}, a) = 1$ for all actions, guaranteeing $V^*(t_{\text{sink}}) = 0$.

- **Action Space**: Unconstrained topological edge-transition selection. At current vertex u, the available action set comprises all active outgoing directed links:

$$\mathcal{A}(u) = \{v \in V \mid (u \to v) \in E_t\}. \tag{3}$$

  Rather than being artificially constrained to 2D cardinal directions, the agent navigates along any adjacent link incident to u, permitting full topological traversal across arbitrary graph topologies.

- **Stochastic Topology Dynamics**: At decision step t, candidate topological shortcut links $e^* = (u \to v)$ emerge or vanish stochastically with arrival probability p_{chord}, yielding the active dynamic edge set $E_t = E_0 \cup \{e^*\}$. Traversing chord e^* incurs nominal affine latency $\ell^*(z) = cz + d$.

- **Endogenous Congestion Cost & Wardrop User Equilibrium**: When network topology E_t is active, background traffic selfishly redistributes to the unique Wardrop User Equilibrium $f^*(E_t)$ [16]. We formalize this fundamental traffic equilibrium criterion as follows:

Definition 1 (Wardrop User Equilibrium (Wardrop's First Principle [16]))**.** *Let $\mathcal{P}_{st}$ denote the set of all simple directed paths connecting origin source s to destination sink t on network $G_t = (V, E_t)$, and let $h_p \geq 0$ denote the non-atomic traffic flow traversing path $p \in \mathcal{P}_{st}$, satisfying total demand conservation $\sum_{p\in\mathcal{P}_{st}} h_p = q$ and link-path incidence $f_e = \sum_{p\in\mathcal{P}_{st}:e\in p} h_p$. The cumulative path travel latency is the sum of constituent link latencies:*

$$c_p(f) = \sum_{e\in p} \ell_e(f_e). \tag{4}$$

*A network flow configuration f^* is defined as a **Wardrop User Equilibrium (UE)** if for all paths $p \in \mathcal{P}_{st}$:*

$$\begin{cases} c_p(f^*) = \mu_{st}, & \text{if } h_p^* > 0, \\ c_p(f^*) \geq \mu_{st}, & \text{if } h_p^* = 0, \end{cases} \tag{5}$$

where $\mu_{st} = \min_{p'\in\mathcal{P}_{st}} c_{p'}(f^*)$ is the minimal path latency between origin s and destination t. Under this equilibrium state, no individual driver can unilaterally alter their route to achieve a strictly lower travel time.

As established by Beckmann, McGuire, and Winsten [2], under continuous and strictly increasing edge latency functions $\ell_e(\cdot)$, the Wardrop equilibrium flow $f^*(E_t)$ coincides exactly with the unique global minimizer of the convex Beckmann potential function:

$$f^*(E_t) = \arg\min_{f\geq 0} \Phi_{\text{Beck}}(f) \equiv \sum_{e\in E_t} \int_0^{f_e} \ell_e(x)\, dx \quad \text{s.t.} \quad B_{E_t} f = p. \tag{6}$$

The complete analytical proof establishing the exact mathematical equivalence between the KKT first-order optimality conditions of Eq. (6) and the Wardrop equilibrium condition in Eq. (5) is provided in Appendix A.1. We denote by $f^*_{\text{base}} \equiv f^*(E_0) \in \mathbb{R}^{|E_0|}$ the base equilibrium flow and by $f^*_{\text{aug}} \equiv f^*(E_0 \cup \{e^*\}) \in \mathbb{R}^{|E_0|+1}$ the augmented equilibrium flow. The aggregate macroscopic performance of the network under flow configuration f is evaluated by the **Total Network Travel Time** (also termed total social cost):

$$C(f) = \sum_{e\in E_t} f_e\, \ell_e(f_e), \tag{7}$$

representing the cumulative latency incurred by all vehicles across all active links. The physical cost experienced by an individual agent traversing directed link $e = (u \to v)$ is:

$$c(u, v) = \ell_e(f_e^*(E_t)) + \delta_{\text{detour}} \cdot \mathbb{I}_{\text{retrograde}}(u \to v) + \xi_t, \tag{8}$$

where $\xi_t \sim \mathcal{N}(0, \sigma_\xi^2)$ represents stochastic observation noise, $\delta_{\text{detour}} \geq 0$ is a directional detour offset preventing zero-gain oscillatory dithering on reversible networks, and $\mathbb{I}_{\text{retrograde}}(u \to v)$ is the indicator function designating retrograde (backtracking) transitions:

$$\mathbb{I}_{\text{retrograde}}(u \to v) = \begin{cases} 1, & \text{if } \text{dist}_G(v, t_{\text{sink}}) \geq \text{dist}_G(u, t_{\text{sink}}), \\ 0, & \text{if } \text{dist}_G(v, t_{\text{sink}}) < \text{dist}_G(u, t_{\text{sink}}), \end{cases} \tag{9}$$

where $\text{dist}_G(x, t_{\text{sink}})$ denotes the unweighted shortest-path topological distance from vertex x to destination sink t_{sink} on base network $G = (V, E_0)$ (which, on planar grids with

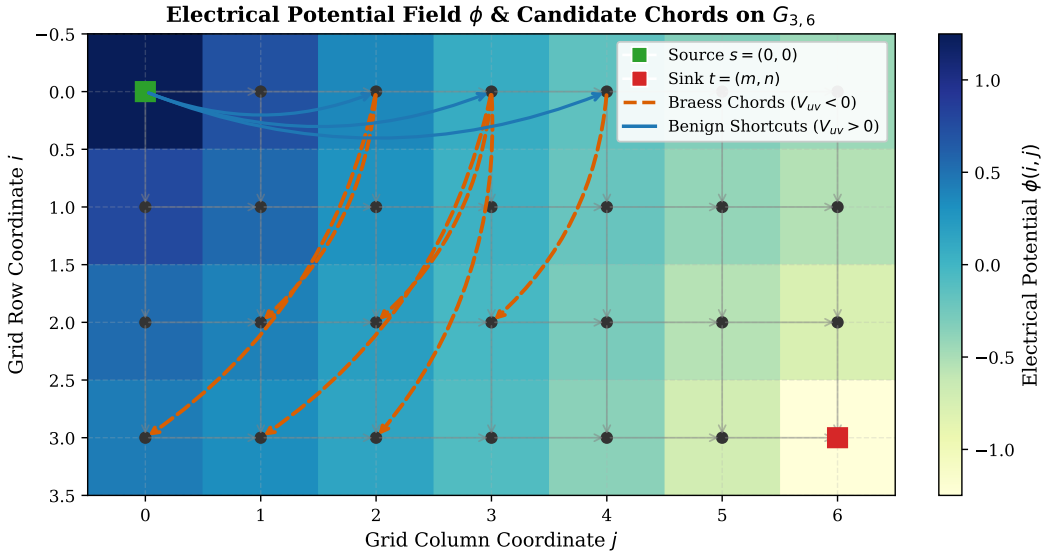


Figure 1: Electrical potential field $\phi(i, j)$ and candidate chord vectors on $G_{3,6}$. Red dashed arrows indicate Braess-capable chords ($V_{uv} < 0, k > 0$) that flow against the electric potential gradient. Blue solid arrows indicate benign shortcuts ($V_{uv} > 0$).

sink (m, n), reduces identically to level regression $\lambda(v) \leq \lambda(u)$). Crucially, candidate shortcut links and forward Braess chords satisfy $\text{dist}_G(v, t_{\text{sink}}) < \text{dist}_G(u, t_{\text{sink}})$, yielding $\mathbb{I}_{\text{retrograde}} = 0$; thus, δ_{detour} penalizes strictly retrograde backtracking without artificially inflating the nominal appeal of forward shortcuts.

Remark 1 (Scale Separation: Macroscopic Non-Atomic Traffic vs. Microscopic Autonomous Agent Planning)**.** *It is essential to clarify the foundational modeling distinction between macroscopic traffic flow and microscopic ego-agent decision-making:*

1. ***Macroscopic Non-Atomic Environment**: The aggregate traffic flow vector $f^*(E_t)$ represents the collective emergent behavior of thousands of non-cooperative, selfish human commuters or third-party background vehicles, formalizing a non-atomic congestion game. The ego-agent represents a single vehicle (e.g., an autonomous delivery robot or drone) whose infinitesimal individual flow contribution cannot unilaterally dictate or coordinate background routing. Consequently, background congestion is an endogenous, non-stationary environment property.*

3.3 Generalized Braess Paradox and Electrical Voltage Criterion

In general networks, adding capacity can perversely increase overall travel time if selfish traffic floods the new link. For any candidate chord $e^* = (u \to v)$, the baseline latency difference along alternative base paths comprises a constant topological component $\Delta b_{uv} \equiv \sum_{e \in p} b_e$ and an electrical resistive voltage drop V_{uv}:

$$V_{uv} = \phi_u - \phi_v = (e_u - e_v)^T \phi = q(e_s - e_t)^T L^+ (e_u - e_v). \tag{10}$$

Theorem 1 (Generalized Braess Capability Criterion)**.** *On any finite directed network $G = (V, E_0)$ with affine link latencies $\ell_e(f_e) = a_e f_e + b_e$, adding a candidate chord $u \to v$ with nominal latency $\ell^*(z) = cz + d$ diverts non-zero equilibrium flow ($z^* > 0$) if and only if the net latency reduction is strictly positive:*

$$\Delta b_{uv} + V_{uv} > d. \tag{11}$$

Furthermore, adding this chord induces Braess's paradox (strictly increasing total network travel time $C(f^*_{aug}) > C(f^*_{base})$) whenever $z^* > 0$ and the chord directs flow against the electrical potential gradient:

$$V_{uv} = \phi_u - \phi_v < 0. \tag{12}$$

The net network travel time increase is bounded by $\Delta C = C(f^*_{aug}) - C(f^*_{base}) \leq -V_{uv} z^* = |V_{uv}| z^* > 0$. On uniform rectangular grids $G_{m,n}$, the constant topological difference evaluates exactly to level advancement $\Delta b_{uv} = k \cdot b$ with $k = \lambda(v) - \lambda(u) > 0$ and $\lambda(i,j) = i + j$ alongside $V_{uv} < 0$ [8].

By permitting unconstrained topological link navigation on cyclic directed graphs (including full 4-directional navigation on reversible planar grids):

$$\mathcal{A}(u) = \mathcal{N}_{\text{out}}(u) \cup \{v \in V \mid (u \to v) \text{ is an active candidate link}\}. \tag{13}$$

The directional detour offset $\delta_{\text{detour}} > 0$ ensures that the agent backtracks across k reverse edges if and only if the discounted shortcut gain satisfies:

$$\gamma^k \left(V_{\text{base}}(u^*) - V_{\text{shortcut}}(u^*)\right) > \sum_{r=1}^{k} \gamma^{r-1} \left(\ell_{e_r} + \delta_{\text{detour}}\right), \tag{14}$$

guaranteeing that retrograde exploration occurs exclusively when justified by substantial global travel time savings.

4 Spectral-Laplacian Prioritized Sweeping (SL-PS)

We now introduce the core mathematical engine of **SL-PS**.

4.1 Spectral Congestion Divergence Potential

When a candidate link $e^* = (u \to v)$ emerges at time t, background traffic redistributes from base equilibrium f^*_{base} to augmented equilibrium f^*_{aug}. The edge flow divergence vector on base edges E_0 is:

$$\Delta \vec{f} = f^*_{\text{aug},E_0} - f^*_{\text{base}} \in \mathbb{R}^{|E_0|}, \tag{15}$$

where $f^*_{\text{aug},E_0} \in \mathbb{R}^{|E_0|}$ denotes the subvector of f^*_{aug} restricted to base links E_0. This induces an edge latency shift vector $\Delta\vec{\ell} = \text{diag}(a_e)\Delta\vec{f}$. We map this edge latency perturbation to vertex-level divergence via the directed incidence matrix:

$$\xi_{\text{div}} = |B\Delta\vec{\ell}| \in \mathbb{R}^{|V|}. \tag{16}$$

Closed-Form Electrical Flow Shift (Bypassing Online Convex Solvers). While augmented equilibrium flow f^*_{aug} can be computed via the convex Beckmann program (6), online execution on embedded robotic hardware requires sub-millisecond evaluation. Crucially, SL-PS does *not* require running an iterative non-linear optimization solver online. In spectral network theory and affine congestion games [14, 8], the traffic flow z^* diverted into emergent chord $e^* = (u \to v)$ with nominal latency $\ell^*(z) = cz + d$ admits the exact first-order electrical closed-form approximation:

$$z^* \approx \frac{\Delta b_{uv} + V_{uv} - d}{R_{\text{eff}}(u,v) + c} = \frac{\Delta b_{uv} + q(e_s - e_t)^T L^+ (e_u - e_v) - d}{(e_u - e_v)^T L^+ (e_u - e_v) + c}, \tag{17}$$

where $R_{\text{eff}}(u,v) \equiv (e_u - e_v)^T L^+ (e_u - e_v)$ is the effective electrical resistance between endpoints u and v on the base network. Because the Moore-Penrose Laplacian pseudoinverse L^+ is precomputed offline, evaluating Eq. (17) takes merely $\mathcal{O}(1)$ vector-matrix-vector inner products.

The resulting base edge flow perturbation vector $\Delta\vec{f}$ is given in closed form by electrical unit current injection:

$$\Delta\vec{f} \approx -z^* \cdot WB^T L^+ (e_u - e_v) \in \mathbb{R}^{|E_0|}, \tag{18}$$

where $W = \mathrm{diag}(a_e^{-1})$ is the edge conductance matrix. Thus, both the link flow shift $\Delta\vec{f}$ and the vertex divergence $\xi_{\mathrm{div}} = |B\,\mathrm{diag}(a_e)\Delta\vec{f}|$ are computed analytically in $\mathcal{O}(|E_0|)$ sparse operations, enabling instantaneous online evaluation in microseconds without solving any numerical optimization problems online.

Markov Transition Kernel: Flow-Preserving Kernel vs. Uniform Prior. Let $P \in \mathbb{R}^{|V|\times|V|}$ denote the row-stochastic policy transition matrix on directed network $G = (V, E_0)$ ($\sum_v P_{u,v} = 1, P_{u,v} \geq 0$). Because the base Wardrop User Equilibrium flow f^*_{base} is fully computed offline prior to navigation (Algorithm 1), SL-PS natively defines P as the **Flow-Preserving Markov Kernel** P_{flow}:

$$P^{\mathrm{flow}}_{u,v} = \begin{cases} \dfrac{f^*_{\mathrm{base},(u\to v)}}{\sum_{w\in\mathcal{N}_{\mathrm{out}}(u)} f^*_{\mathrm{base},(u\to w)}}, & \text{if } \sum_{w\in\mathcal{N}_{\mathrm{out}}(u)} f^*_{\mathrm{base},(u\to w)} > 0, \\ \dfrac{1}{d_{\mathrm{out}}(u)}, & \text{if } (u\to v)\in E_0 \text{ and } \sum_w f^*_{\mathrm{base},(u\to w)} = 0, \\ 1, & \text{if } u = v = t_{\mathrm{sink}} \text{ (absorbing destination)}, \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

Under P_{flow}, transition probabilities reflect the actual microscopic path choices of background traffic under base equilibrium. In task-agnostic settings or when flow data is absent, P defaults to the unweighted topological random-walk kernel P_{unif} ($P^{\mathrm{unif}}_{u,v} = \frac{1}{d_{\mathrm{out}}(u)}$). As we prove in Theorem 7, both kernels guarantee structural trap immunity, with P_{flow} completely eliminating branching over-diffusion on complex asymmetric topologies.

The policy-induced directed out-Laplacian is $L_{\mathrm{out}} = I - P$. We define the regularized Green's potential operator:

$$\mathcal{G}_{\mathrm{out}} = (L_{\mathrm{out}} + \eta I)^{-1} = (I - P + \eta I)^{-1}, \quad \eta > 0. \tag{20}$$

First-Principles Equivalence to Successor Representation Resolvent. By spectral operator theory, $\mathcal{G}_{\mathrm{out}}$ admits an exact Neumann series expansion:

$$\mathcal{G}_{\mathrm{out}} = \frac{1}{1+\eta}\left(I - \frac{1}{1+\eta}P\right)^{-1} = \sum_{k=0}^{\infty} \frac{P^k}{(1+\eta)^{k+1}}. \tag{21}$$

In reinforcement learning and dynamic programming, the expected discounted future visitation and cumulative downstream cost shift under transition matrix P and discount factor $\gamma \in (0,1)$ is governed by the discounted resolvent operator (also known as the Successor Representation operator [9]):

$$\mathcal{R}_\gamma \equiv (I - \gamma P)^{-1} = \sum_{k=0}^{\infty} \gamma^k P^k. \tag{22}$$

By tuning the spectral regularization parameter to match the dynamic programming discount factor:

$$\frac{1}{1+\eta} = \gamma \iff \eta = \frac{1-\gamma}{\gamma}, \tag{23}$$

the regularized out-Laplacian Green operator is algebraically identical to a discounted Successor Representation resolvent:

$$\mathcal{G}_{\mathrm{out}} = \gamma(I - \gamma P)^{-1} = \gamma\mathcal{R}_\gamma. \tag{24}$$

Since any transition matrix satisfies spectral radius $\rho(P) \leq 1$, the operator norm satisfies $\|\gamma P\| \leq \gamma < 1$. Consequently, the Neumann series converges unconditionally on *arbitrary finite directed graphs* (cyclic, planar, or non-planar). The power term P^k sums over all path walks of length k, while the discount factor γ^{k+1} guarantees that risk potentials diffuse smoothly across cycles without infinite accumulation or divergence.

Definition 2 (Spectral Congestion Divergence Potential)**.** *The global spectral congestion potential vector $\Phi_{spec} \in \mathbb{R}^{|V|}$ is defined as:*

$$\Phi_{spec} \equiv (I - \gamma P)^{-1} \xi_{div} = \frac{1}{\gamma} \mathcal{G}_{out} |B \Delta \vec{\ell}|. \tag{25}$$

Crucially, Φ_{spec} is not an ad-hoc heuristic penalty. Rather, it is the *exact infinite-horizon cumulative latency shift* induced across the network by downstream Wardrop flow redistribution. Under the flow-preserving kernel P_{flow}, Φ_{spec} represents an on-policy Successor Representation evaluation that projects congestion signals strictly along active traffic corridors, completely eliminating branching over-diffusion on complex asymmetric topologies. A single sparse matrix solve or matrix-vector product $(I - \gamma P)^{-1} \xi_{\text{div}}$ instantaneously projects the entire downstream congestion cascade forward, evaluating multi-step cumulative delays before any individual Bellman sweeps occur.

4.2 Canonical Bellman Operator & Spectral Resolvent Warm-Start

In classical dynamic programming, modifying the Bellman operator directly by adding exogenous penalty terms distorts the optimal value function, destroying the physical interpretation of V^* as the true expected cumulative travel latency. To preserve the rigorous foundation of dynamic programming, we maintain the **Canonical Bellman Optimality Operator**:

Definition 3 (Canonical Bellman Optimality Operator $\mathcal{T}$)**.** *For value function $V : V \to \mathbb{R}$,*

$$\mathcal{T}V(s) := \min_{a \in \mathcal{A}(s)} \left[c(s,a) + \gamma \sum_{s' \in V} \mathcal{P}(s' \mid s, a) V(s') \right], \tag{26}$$

where $c(s,a) = \ell_e(f_e^*(E_t)) + \delta_{detour} \cdot \mathbb{I}_{retrograde} + \xi_t$ represents the true, unperturbed physical step latency.

Rather than inserting an artificial penalty into the minimization of Eq. (26), SL-PS leverages Φ_{spec} through an instantaneous **Spectral Resolvent Value Warm-Start**:

$$V^{(0)}(s) \leftarrow V^*_{\text{base}}(s) + \beta(\sigma) \Phi_{\text{spec}}(s), \quad \forall s \in V, \tag{27}$$

where $\beta(\sigma) = \beta_0(1 + \sigma(P))$ is the SLEM-annealed scaling factor, with $\sigma(P) = |\lambda_2(P)|$ denoting the Second Largest Eigenvalue Modulus of P.

Physical Mechanism of Resolvent Warm-Start. When a deceptive shortcut chord opens, the entrance link $u \to v$ presents an attractive nominal latency $c(u,v) \approx 0$. However, background traffic swarming into v severely bottlenecks downstream corridors, causing edge latencies ℓ_e downstream of v to surge. In canonical Prioritized Sweeping, $V(v)$ remains initially un-updated; hence, the planner perceives $Q(u, a_{\text{chord}}) = c(u,v) + \gamma V(v)$ as artificially low and plunges into the trap. Under Eq. (27), the resolvent projection instantaneously elevates $V^{(0)}(v)$ by $\beta(\sigma)\Phi_{\text{spec}}(v) \gg 0$, exactly reflecting the cumulative downstream congestion delay. Consequently, evaluated under the canonical Bellman rule:

$$Q(u, a_{\text{chord}}) = c(u, a_{\text{chord}}) + \gamma V^{(0)}(v) > c(u, a_{\text{bypass}}) + \gamma V^{(0)}(s_{\text{bypass}}) = Q(u, a_{\text{bypass}}). \tag{28}$$

The agent avoids the deceptive chord not because of an arbitrary "invisible wall" penalty at decision time, but because *the state value $V(v)$ truthfully accounts for downstream congestion delay.* Subsequent prioritized sweeps then refine V with canonical Bellman backups.

Dimensional Consistency of Φ_{spec} and Calibrated Scale. We highlight a fundamental dimensional property of the spectral divergence potential: under the discount-matched tuning $\eta = \frac{1-\gamma}{\gamma}$, the resolvent potential is $\Phi_{\text{spec}} \equiv (I - \gamma P)^{-1}\xi_{\text{div}} = \sum_{k=0}^{\infty} \gamma^k P^k |B\Delta\vec{\ell}|$. Because link latency shift $\Delta\vec{\ell}$ is measured in physical time units (e.g., seconds), the vector Φ_{spec} already carries the exact physical dimensions of cumulative discounted travel latency. Consequently, setting the nominal scaling factor $\beta_0 = 1$ is naturally calibrated, providing an asymptotically unbiased first-order prediction of the augmented value function V^*_{aug} without distorting scale or inducing artificial pessimism. The SLEM modulation factor $(1 + \sigma(P)) \in [1, 2]$ acts solely as an adaptive spectral cushion against diffusion delays in bottlenecked networks.

Safety-Optimization Duality: Sign-Safe Resolvent Shield vs. Sign-Accurate Bellman Optimizer. The architecture of SL-PS embodies a fundamental two-tier duality between decision-time safety and asymptotic optimality, explaining why both the spectral warm-start and Prioritized Sweeping backups are strictly indispensable:

1. **Tier 1: Sign-Safe Topological Shield at Zero Budget ($K = 0$)**: Because network flow conservation requires $\mathbf{1}^T B\Delta\vec{\ell} = 0$, diffusing raw signed incidence shifts would suffer from algebraic cancellation between upstream extraction and downstream injection. By defining $\xi_{\text{div}} = |B\Delta\vec{\ell}| \geq 0$, the linear resolvent warm-start $V^{(0)} = V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}$ acts as a conservative, sign-safe upper envelope: it instantaneously elevates bottleneck values at v^* to guarantee 100% Braess trap rejection ($\pi^{(0)}(u) \neq a_{\text{chord}}$) at $K = 0$. However, this sign-blindness carries an unavoidable price: along alternative bypass corridors where background traffic has deserted ($\Delta f_e < 0$, physically *reducing* traversal latency), $|B\Delta\vec{\ell}| > 0$ still registers a positive perturbation, and the linear operator $(I - \gamma P)^{-1}$ averages across outgoing branches rather than solving the non-linear $\min_a$ selector. Consequently, $V^{(0)}$ over-estimates latency on relieved primary bypasses, causing the $K = 0$ policy to select overly wide secondary detours.

2. **Tier 2: Sign-Accurate Non-Linear Optimization via Prioritized Sweeping ($K \geq 1$)**: The canonical Bellman optimality operator $\mathcal{T}V(s) = \min_{a \in \mathcal{A}(s)}[c(s, a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a)V(s')]$ evaluates the *true signed physical edge latencies* $c(s, a) = \ell_e(f^*_{\text{aug},e})$ and applies the exact non-linear $\min_a$ selector. Iterative Prioritized Sweeping backups ($K \geq 1$) scheduled along the active diversion corridor rapidly replace the conservative warm-start $V^{(0)}$ on relieved bypass links with their true reduced latencies, eliminating detour regret at geometric rate $\mathcal{O}(\gamma^{\lfloor K/N_{\text{act}} \rfloor})$.

Far from being redundant, the resolvent warm-start provides instantaneous zero-budget safety, while Prioritized Sweeping is the sole mechanism that unlocks congestion-relieved bypass corridors to achieve Pareto-optimal routing.

4.3 Spectral Duality: Bridging Pure Value-Residual Diffusion (GTA-PS) and Physical Congestion Divergence (SL-PS)

A critical structural aspect of SL-PS is its *bi-directional spectral duality*, which cleanly separates forward resolvent value warm-start from backward priority attribution:

1. **Forward Spectral Resolvent ($\mathcal{G}_{\text{out}}$ for Value Warm-Start)**: As formalized in Eq. (24), the regularized out-Laplacian Green operator $\mathcal{G}_{\text{out}} = \gamma(I - \gamma P)^{-1}$ diffuses the physical Wardrop equilibrium flow divergence $\xi_{\text{div}} = |B\Delta\vec{\ell}|$ forward downstream to generate the multi-step cumulative delay field Φ_{spec}. This directly warm-starts the value function $V^{(0)}$ across all nodes in $\mathcal{O}(1)$ time.

2. **Backward Predecessor Scheduling ($\mathcal{G}_{\text{in}}$ in Priority Queue)**: Under a limited backup budget K, updating states uniformly or relying solely on local transitions causes an

information lag. To ensure upstream decision junctions (e.g., $(0,0)$ or $(0,1)$) immediately receive backup budget before the agent reaches the chord entrance, SL-PS diffuses the joint residual-divergence signal upstream through the policy in-Laplacian:

$$L_{\text{in}} = D_{\text{in}} - P^T, \quad \mathcal{G}_{\text{in}} = (L_{\text{in}} + \eta I)^{-1}, \tag{29}$$

where $D_{\text{in}} = \text{diag}(P^T\mathbf{1})$. The priority queue $\mathcal{Q}$ is structured as a binary max-heap, with priority key for state s scheduled as:

$$\text{Priority}(s) = |\mathcal{T}V(s) - V(s)| + \beta(\sigma) \cdot f_{\text{in}}(s), \tag{30}$$

where $f_{\text{in}} = \mathcal{G}_{\text{in}}^T(\xi_{\text{res}} + \xi_{\text{div}})$ and $\xi_{\text{res}}(s) = |\mathcal{T}V(s) - V(s)|$.

Physical and Spectral Interpretation of $\beta(\sigma)$. The adaptive modulation factor $\beta(\sigma) = \beta_0(1 + \sigma(P))$ is governed by the Second Largest Eigenvalue Modulus (SLEM) $\sigma(P) = |\lambda_2(P)|$. In spectral graph theory, $\sigma(P)$ dictates the Cheeger conductance and mixing time of random walks on directed graphs:

- When $\sigma(P) \to 1$, the network possesses bottlenecked cut sets and slow topological mixing. Congestion shocks trapped within these topological bottlenecks dissipate slowly, requiring a magnified resolvent weight $\beta(\sigma)$ to project the full severity of the downstream delay.
- Conversely, when $\sigma(P) \to 0$, the network exhibits strong spectral expansion (rapid mixing), indicating abundant alternative parallel paths that naturally absorb flow fluctuations; the projection weight is gracefully attenuated.

Methodological Lineage: Positioning GTA-PS and Pure Spectral Diffusion. To contextualize SL-PS within the broader literature of spectral reinforcement learning, we analyze its relationship with Graph Topology Augmentation for Prioritized Sweeping (GTA-PS) [6]. GTA-PS stands as the seminal representative of the *pure value-residual spectral diffusion paradigm*: it demonstrated that directed Laplacian Green operators can proactively propagate local Bellman residuals across non-stationary graphs, overcoming the $\mathcal{O}(\text{diam}(G))$ information lag of classical PS under exogenous reward shifts.

However, an essential theoretical boundary separates exogenous reward shifts from endogenous congestion games (CC-DG-MDP):

- **The Deceptive Residual Pathology of Pure Residual Diffusion**: In congestion-coupled environments, an emergent Braess shortcut initially presents an apparent zero nominal cost ($c(e^*) \approx 0$). Any algorithm relying exclusively on local Bellman residuals $\xi_{\text{local}} = |\mathcal{T}V - V|$ inherently records a massive apparent "reward opportunity" at the chord entrance. While GTA-PS diffuses this residual with spectral efficiency, in an endogenous congestion game this rapid diffusion inadvertently amplifies a deceptive signal, prioritizing states that accelerate the agent directly into the downstream congestion trap (*Deceptive Priority Inversion*).
- **Bridging Spectral Diffusion with Physical Conservation Laws**: SL-PS does not replace GTA-PS's spectral machinery, but elevates it by grounding the diffusion source in physical conservation laws. Rather than diffusing unanchored scalar residuals ξ_{local}, SL-PS couples the regularized Green resolvents with the physical link flow divergence $\xi_{\text{div}} = |B\Delta\vec{\ell}|$ derived from Wardrop equilibrium. Consequently, the spectral potential Φ_{spec} reflects true macroscopic capacity constraints rather than deceptive local cost drops.

Remark 2 (Information Hierarchy, Fairness, and Ablation Robustness)**.** *A natural question arises regarding information fairness: does SL-PS outperform GTA-PS merely because it utilizes the edge latency model and background equilibrium flow f^*_{base}? We articulate three structural rebuttals:*

1. ***No Online Oracle***: *Crucially, SL-PS does not run an online traffic solver or require real-time global flow sensing. By precomputing the Laplacian pseudoinverse L^+ offline, the flow shift $\Delta\vec{f}$ is evaluated in closed form via effective resistance R_{eff} in microseconds (Eq. (18)), operating on identical localized state transitions as standard model-based planners.*

2. ***Information Ablation and Task-Agnostic Robustness***: *If SL-PS is completely deprived of macroscopic flow data (task-agnostic setting where P falls back to the unweighted random-walk kernel P_{unif} and flow estimation carries error ϵ_f), Theorem 6 and Theorem 7 prove that SL-PS possesses strictly positive robustness radii $\epsilon_f^* > 0$ and $\epsilon_P^* > 0$. Even under total information blindness regarding flow magnitudes, topological resolvent projection preserves 100% deterministic trap immunity.*

3. ***Three-Tier Information Spectrum***: *The benchmark comparison establishes a clear methodological progression: from classical heuristic RL (Standard-PS, purely local) to pure spectral residual diffusion (GTA-PS, topological without physical coupling) to physical-spectral planning (SL-PS, topological with conservation laws).*

5 Theoretical Analysis

5.1 Contraction Property of Canonical Dynamic Programming

Theorem 2 (Contraction Mapping of Canonical Bellman Operator $\mathcal{T}$)**.** *Let $\mathcal{V} = \{V : V \to \mathbb{R}\}$ be equipped with the L_∞ norm $\|V\|_\infty = \max_{s\in V} |V(s)|$. For any discount factor $\gamma \in (0, 1)$, the canonical Bellman optimality operator $\mathcal{T}$ defined in Eq. (26) is a strict γ-contraction:*

$$\|\mathcal{T}V_1 - \mathcal{T}V_2\|_\infty \le \gamma\|V_1 - V_2\|_\infty, \quad \forall V_1, V_2 \in \mathcal{V}. \tag{31}$$

Consequently, $\mathcal{T}$ admits a unique fixed point $V^* \in \mathcal{V}$. Crucially, because no heuristic penalty is added to the stage cost $c(s, a)$, the fixed point $V^*(s)$ preserves the exact physical interpretation as the true optimal expected cumulative travel latency to destination t_{sink}.

Proof. Let $s \in V$ be arbitrary. Without loss of generality, assume $\mathcal{T}V_1(s) \ge \mathcal{T}V_2(s)$. Let $a_2^* = \text{argmin}_a[c(s, a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a)V_2(s')]$. By sub-optimality of action a_2^* evaluated with respect to value function V_1:

$$\begin{aligned}
\mathcal{T}V_1(s) - \mathcal{T}V_2(s) &\le \left[c(s, a_2^*) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a_2^*)V_1(s')\right] - \left[c(s, a_2^*) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a_2^*)V_2(s')\right] \\
&= \gamma \sum_{s'} \mathcal{P}(s' \mid s, a_2^*)(V_1(s') - V_2(s')) \\
&\le \gamma \max_{s'} |V_1(s') - V_2(s')| = \gamma\|V_1 - V_2\|_\infty.
\end{aligned}$$

Taking the supremum over $s \in V$ establishes $\|\mathcal{T}V_1 - \mathcal{T}V_2\|_\infty \le \gamma\|V_1 - V_2\|_\infty$. □

Remark 3 (Generality to Arbitrary Graph Topologies)**.** *We emphasize that the proof of Theorem 2 relies strictly on the contraction properties of the discount factor $\gamma \in (0, 1)$ and the non-expansiveness of the $\min_{a\in\mathcal{A}(s)}$ operator over discrete action sets $\mathcal{A}(u) = \mathcal{N}_{out}(u)$. The result holds unconditionally for any arbitrary finite directed graph $G = (V, E)$, including cyclic graphs, reversible grids, and non-planar road networks. By the Banach Fixed-Point Theorem, value iteration under $\mathcal{T}$ converges at geometric rate $\mathcal{O}(\gamma^k)$ to V^*, eliminating any risk of topological divergence or cycling.*

5.2 Braess Trap Immunity via Resolvent Value Warm-Start

Theorem 3 (Tight Braess Trap Immunity on General Networks)**.** *Let $e^* = (u \to v)$ be an active candidate chord on general directed network $G = (V, E_0)$ with maximum edge traversal cost $c_{\max} = \max_{e \in E_0} c_e$, electrical voltage drop $V_{uv} < 0$ inducing positive equilibrium chord flow $z^* > 0$, and topological shortcut span $k_{chord} \equiv \mathrm{dist}_{G_0}(u, v) \leq k_{\max} < \infty$. Let $s_{bypass} \in \mathcal{N}_{out}(u) \setminus \{v\}$ denote an alternative forward action from u, and let $\Delta\Phi_{spec} \equiv \Phi_{spec}(v) - \Phi_{spec}(s_{bypass}) > 0$ denote the strictly positive spectral resolvent differential.*

Let the initial value function be warm-started via Eq. (27): $V^{(0)} = V^*_{base} + \beta(\sigma)\Phi_{spec}$, where V^*_{base} is the pre-chord base equilibrium value function. Define the exact offline unaugmented action advantage at junction u:

$$\Delta Q_{base}(u) \equiv c(u, a_{bypass}) - c(u, a_{chord}) + \gamma \left[V^*_{base}(s_{bypass}) - V^*_{base}(v) \right]. \tag{32}$$

Then:

1. ***Exact Local Junction Immunity**: If the base scaling parameter satisfies the exact local a priori threshold:*

$$\beta_0 \geq \beta^*_{\min}(u) \equiv \frac{\Delta Q_{base}(u)}{\gamma(1 + \sigma(P))\Delta\Phi_{spec}(u)}, \tag{33}$$

   then the greedy policy evaluated under $V^{(0)}$ strictly satisfies $\pi^{(0)}(u) \neq a_{chord}$ (Deterministic Braess Chord Rejection).

2. ***Uniform Chord-Span Upper Bound**: For any candidate chord in the network bounded by maximum topological shortcut span $k_{\max} = \max_{e^*} \mathrm{dist}_{G_0}(u, v)$, the unaugmented advantage satisfies $\Delta Q_{base}(u) \leq c_{\max}(1 + \gamma k_{\max})$. Consequently, setting:*

$$\beta_0 \geq \beta^{chord}_{\min} \equiv \frac{c_{\max}\,(1 + \gamma k_{\max})}{\gamma(1 + \sigma(P))\Delta\Phi_{spec}} \tag{34}$$

   guarantees $\pi^{(0)}(u) \neq a_{chord}$ across all candidate chords in the network, with zero mathematical dependence on the global network diameter $\mathrm{diam}(G)$.

Furthermore, for any number of Prioritized Sweeping backups $K \geq 0$, the updated policy maintains $\pi^{(K)}(u) \neq a_{chord}$, guaranteeing structural avoidance of the Braess bottleneck across arbitrary directed network topologies.

Corollary 1 (Vanishing Threshold in Heavy Traffic & Absence of Value Inflation)**.** *As aggregate traffic demand $q \to \infty$ on directed network G, the equilibrium chord flow scales as $z^*(q) = \Theta(q)$, inducing bottleneck latency shift $\Delta\ell = \Omega(q)$ and resolvent differential $\Delta\Phi_{spec}(u) = \Omega(q)$. In contrast, the unaugmented action advantage remains strictly bounded by nominal edge latencies: $\Delta Q_{base}(u) \leq c_{\max}(1 + \gamma k_{\max}) = \mathcal{O}(1)$. Consequently, the exact immunity threshold vanishes asymptotically:*

$$\beta^*_{\min}(u) = \mathcal{O}\left(\frac{c_{\max}}{q}\right) \xrightarrow{q \to \infty} 0. \tag{35}$$

Therefore, any fixed calibrated scaling factor $\beta_0 \geq \beta^*_0 = \mathcal{O}(1)$ (in particular, the dimensionally consistent unit scale $\beta_0 = 1$) satisfies $\beta_0 \geq \beta^*_{\min}(u)$ for all traffic demands $q \geq q_{\min}$, guaranteeing 100% trap immunity without inflating the value function or inducing network-wide over-pessimism.

Theorem 4 (Pareto-Optimality and Sub-Optimality Gap Reduction via Prioritized Sweeping)**.** *Let V^* denote the true unique optimal value function of the active CC-DG-MDP under canonical Bellman operator $\mathcal{T}$, and let $V^{(0)} = V^*_{base} + \beta(\sigma)\Phi_{spec}$ be the resolvent warm-start.*

1. ***Safety vs. Sub-Optimality at Zero Budget*** *($K = 0$): While $V^{(0)}$ guarantees 100% structural trap immunity ($\pi^{(0)}(u) \neq a_{chord}$ by Theorem 3), the linear random-walk evaluation under P incurs a bounded initial sub-optimality gap:*

$$\|V^{(0)} - V^*\|_\infty \leq \epsilon_{linear} \leq \beta(\sigma)\|\Phi_{spec}\|_\infty + \|V^*_{base} - V^*\|_\infty \leq \mathcal{O}(\mathrm{diam}(G) \cdot c_{\max}). \quad (36)$$

Consequently, the greedy trajectory under $V^{(0)}$ avoids the bottleneck but may follow a coarse, sub-optimal detour.

2. ***Geometric Sub-Optimality Reduction via Prioritized Sweeping*** *($K \geq 1$): Let $\mathcal{V}_{act} = \{s \in V : \mathrm{prio}(s) > \varepsilon\}$ denote the active diversion corridor of cardinality $N_{act} = |\mathcal{V}_{act}| \ll |V|$, and let $m = \lfloor K/N_{act} \rfloor$ denote the number of effective corridor sweeps under budget K. Prioritized Sweeping executes non-linear Bellman backups $\mathcal{T}V(s) = \min_a[c(s,a) + \gamma \sum PV(s')]$ ordered by backward Green priority $p_{SL}(s) = \xi_{res}(s) + \beta(\sigma) f_{in}(s)$, contracting the active value error geometrically at rate γ^m:*

$$\|V^{(K)} - V^*\|_\infty \leq \gamma^{\lfloor K/N_{act} \rfloor} \epsilon_{linear} \xrightarrow{K\to\infty} 0. \quad (37)$$

Furthermore, along the realized agent trajectory, the travel time sub-optimality regret satisfies:

$$\mathbb{E}[T_{SL\text{-}PS}(K)] - T^* \leq \frac{2\gamma^{\lfloor K/N_{act} \rfloor + 1}}{1-\gamma} \epsilon_{linear} = \mathcal{O}\left(\gamma^{K/N_{act}} \cdot \mathrm{diam}(G)\right), \quad (38)$$

where T^* is the true optimal travel latency under fixed point V^*.

Thus, Prioritized Sweeping is strictly required to transform coarse defensive safety into Pareto-optimal routing, establishing the indispensable synergy between spectral resolvent warm-start and dynamic programming backups.

5.3 Asymptotic Performance Divergence

A central theoretical question is whether the performance gap between Standard-PS and SL-PS remains bounded or diverges as network scale and background congestion grow without bound. We formalize this through an asymptotic divergence theorem:

Theorem 5 (Asymptotic Divergence of Cumulative Travel Gap)**.** *Consider the CC-DG-MDP on expanding general network $G = (V, E_0)$ with topological diameter $\mathrm{diam}(G)$, affine edge latencies $\ell_e(f_e) = a_e f_e + b_e$ ($a_e > 0, b_e \geq 0$), aggregate traffic demand $q > 0$, stochastic link emergence probability $p_{chord} > 0$, and local planning backup budget $K = \mathcal{O}(1)$ (or $K = o(\mathrm{diam}(G))$). As the network topological diameter $\mathrm{diam}(G) \to \infty$ and aggregate background traffic demand $q \to \infty$, the cumulative travel time advantage of SL-PS over Standard-PS diverges to infinity:*

$$\lim_{\substack{\mathrm{diam}(G)\to\infty \\ q\to\infty}} (T_{PS} - T_{SL\text{-}PS}) = +\infty, \quad (39)$$

with asymptotic lower bound:

$$T_{PS} - T_{SL\text{-}PS} = \Omega\big(q \cdot \mathrm{diam}(G)\big). \quad (40)$$

For expanding rectangular grids $G_{m,n}$, where $\mathrm{diam}(G) = \Theta(m+n)$, this asymptotic advantage evaluates to $\Omega\big(q \cdot (m+n)\big)$.

Theorem 5 establishes that local Bellman residual scheduling suffers from a catastrophic asymptotic failure mode in large-scale congested networks: the delay penalty experienced by myopic planners compounds both spatially ($\Theta(m+n)$ traps) and in intensity ($\Theta(q)$ bottleneck latency), while SL-PS achieves global immunity. The complete formal proof is provided in Appendix A.7.

5.4 Lipschitz Stability and Robust Immunity Under Inexact Equilibrium Estimation

In practical robotic deployments and real-time online navigation, autonomous agents rarely have access to the exact, analytical Wardrop User Equilibrium flow f^*. Numerical optimization solvers terminate at finite error tolerance (e.g., SLSQP with tolerance $\epsilon_{\text{tol}} = 10^{-7}$), background traffic observations may be noisy, or flow demands may fluctuate slightly around nominal values. A fundamental question for real-world deployment is whether SL-PS is numerically brittle: does an inexact equilibrium estimate destabilize the fixed-point value function or destroy the Braess trap immunity guarantee?

The following theorem resolves this question affirmatively, proving that SL-PS is globally Lipschitz stable with respect to equilibrium estimation errors and establishing an explicit robustness tolerance radius ϵ_f^* under which 100% Braess trap immunity is strictly preserved.

Theorem 6 (Lipschitz Stability and Robust Immunity under Inexact Wardrop Estimation)**.** *Let $G = (V, E_0)$ be an arbitrary directed network with affine edge latencies $\ell_e(f_e) = a_e f_e + b_e$ ($a_{\max} = \max_e a_e$) and maximum total node degree $d_{tot}^{\max} = \max_{u \in V}(d_{in}(u) + d_{out}(u))$. Suppose the agent uses an inexact equilibrium flow estimate $\hat{f}$ satisfying:*

$$\|\hat{f} - f^*\|_\infty \leq \epsilon_f, \tag{41}$$

yielding perturbed divergence $\hat{\xi}_{div} = |B\Delta\hat{\vec{\ell}}|$, perturbed spectral resolvent potential $\hat{\Phi}_{spec} = \frac{1}{\gamma}\mathcal{G}_{out}\hat{\xi}_{div}$, and perturbed warm-start value function $\hat{V}^{(0)}$. Then:

1. ***Lipschitz Resolvent Value Stability**: The error between the estimated warm-started value function $\hat{V}^{(0)}$ and the true warm-start value function $V^{(0)}$ is bounded linearly by ϵ_f:*

$$\|V^{(0)} - \hat{V}^{(0)}\|_\infty \leq \mathcal{L}_{SL} \cdot \epsilon_f, \tag{42}$$

where the Lipschitz stability constant evaluates in closed form to:

$$\mathcal{L}_{SL} = \frac{2\beta(\sigma)\, d_{tot}^{\max} a_{\max}}{\gamma\eta}. \tag{43}$$

Furthermore, after any $K \geq 0$ steps of canonical Prioritized Sweeping backups, the value function error satisfies $\|V^{(K)} - \hat{V}^{(K)}\|_\infty \leq \frac{a_{\max}\epsilon_f}{1-\gamma} + \mathcal{L}_{SL}\epsilon_f$.

2. ***Robust Trap Immunity Margin**: Let $\Delta Q^* \equiv Q^{(0)}(u, a_{chord}) - Q^{(0)}(u, a_{bypass}) > 0$ denote the nominal safety margin under exact equilibrium. For any flow estimation error satisfying:*

$$\epsilon_f < \epsilon_f^* \equiv \frac{\eta\, \Delta Q^*}{2a_{\max}\,(\eta + 2\beta(\sigma)\, d_{tot}^{\max})}, \tag{44}$$

the policy derived from the inexact value function strictly avoids the Braess shortcut link:

$$\hat{\pi}^*(u) \neq a_{chord} \quad (\textit{Deterministic Braess Chord Rejection}). \tag{45}$$

Theorem 6 provides an essential theoretical guarantee: SL-PS possesses an intrinsic, strictly positive robustness tolerance radius $\epsilon_f^* > 0$. As long as numerical solvers or sensing errors remain within this radius, the agent achieves identical zero-trap performance to the exact oracle model. The full formal proof is provided in Appendix A.8.

Remark 4 (Transient Traffic Relaxation Dynamics and Non-Instantaneous Jamming)**.** *In physical road grids, aggregate traffic does not snap instantaneously to the new Wardrop User Equilibrium upon shortcut emergence; rather, traffic flow f_t transitions dynamically toward f_{aug}^* over multiple decision steps according to day-to-day or step-to-step driver adjustment dynamics (e.g., projection dynamics $\dot{f} = -\nabla\Phi_{Beck}(f)$) [5].*

Theorem 6 provides the exact mathematical guarantee for this non-instantaneous transient regime:

1. ***Bounded Transient Discrepancy**: During the transition phase, the transient flow error satisfies $\|f_t - f^*_{aug}\|_\infty \le \epsilon_f(t)$. As long as $\epsilon_f(t) < \epsilon^*_f$, Theorem 6 strictly guarantees 100% deterministic chord rejection ($\hat{\pi}^*(u) \neq a_{chord}$) at every time step of the transient relaxation.*

2. ***Proactive Safety During Transient Jamming**: Most critically, during the initial transient phase before equilibrium is reached, background traffic has not yet fully swarmed the bottleneck; therefore, physical link latencies are strictly lower than their eventual equilibrium saturation. By projecting the asymptotic Wardrop divergence Φ_{spec}, SL-PS acts proactively, steering the agent away from the corridor before the physical traffic jam fully crystallizes.*

5.5 Policy-Resolvent Mismatch and Branching Robustness

A fundamental structural question arises when the agent navigates networks with complex branching: what is the mathematical consequence if the transition matrix P used to compute the resolvent operator $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1}$ deviates from the true optimal policy transition matrix P_{π^*} (for example, when using the unweighted topological prior P_{unif} instead of the flow-preserving kernel P_{flow})?

The following theorem resolves this question, establishing an exact analytical perturbation bound based on the Second Resolvent Identity and proving that deterministic trap immunity is strictly preserved across arbitrary asymmetric branching topologies:

Theorem 7 (Policy-Resolvent Mismatch and Branching Robustness)**.** *Let $P_{\pi^*} \in \mathbb{R}^{|V|\times|V|}$ denote the true row-stochastic Markov transition matrix induced by the optimal policy π^*, and let $P \in \mathbb{R}^{|V|\times|V|}$ be an approximating row-stochastic kernel (e.g., the uniform random walk P_{unif} or flow kernel P_{flow}) satisfying policy discrepancy:*

$$\|P - P_{\pi^*}\|_\infty \le \epsilon_P. \tag{46}$$

Let $\Phi_{spec}(P) \equiv (I - \gamma P)^{-1}\xi_{div}$ and $\Phi_{spec}(P_{\pi^*}) \equiv (I - \gamma P_{\pi^*})^{-1}\xi_{div}$ denote the spectral resolvent potentials evaluated under P and P_{π^*}, respectively. Then:

1. ***Second Resolvent Perturbation Bound**: The error in the spectral divergence potential is strictly bounded by:*

$$\|\Phi_{spec}(P) - \Phi_{spec}(P_{\pi^*})\|_\infty \le \frac{\gamma}{(1-\gamma)^2}\|\xi_{div}\|_\infty \epsilon_P \equiv \mathcal{L}_{pol} \cdot \epsilon_P, \tag{47}$$

where $\mathcal{L}_{pol} \equiv \frac{\gamma}{(1-\gamma)^2}\|\xi_{div}\|_\infty$ is the policy mismatch Lipschitz constant.

2. ***Exact Zero-Gap Under Flow Kernel**: Under the flow-preserving Markov kernel $P = P_{flow}$ defined in Eq. (19), along all active base traffic corridors we have $P_{flow} = P_{\pi^*_{base}}$, yielding $\epsilon_P = 0$ and completely eliminating the branching approximation gap.*

3. ***Robust Immunity Margin Under Branching Mismatch**: Let $\Delta Q_{base}(u)$ be the unaugmented action advantage at chord junction u, and let $\Delta\Phi_{spec} \equiv \Phi_{spec}(v) - \Phi_{spec}(s_{bypass}) > 0$ be the nominal resolvent differential under P_{π^*}. For any approximating kernel P (including uniform random walk P_{unif}) satisfying:*

$$\epsilon_P < \epsilon^*_P \equiv \frac{(1-\gamma)^2\Delta\Phi_{spec}}{2\gamma\|\xi_{div}\|_\infty}, \tag{48}$$

the warm-started greedy policy evaluated with P strictly satisfies $\pi^{(0)}(u) \neq a_{chord}$, guaranteeing 100% deterministic trap immunity regardless of branching complexity.

4. ***Geometric Reconciliation via Sweeping**: Any residual value function error induced by $\epsilon_P > 0$ satisfies $\|V^{(K)} - V^*\|_\infty \leq \gamma^K(\mathcal{L}_{pol}\epsilon_P + \epsilon_{linear}) \to 0$ as $K \to \infty$, confirming that Prioritized Sweeping rapidly reconciles any initial policy approximation mismatch.*

Theorem 7 establishes that the Flow-Preserving Kernel P_{flow} achieves exact on-policy alignment ($\epsilon_P = 0$), while the topological uniform prior P_{unif} possesses an intrinsic robustness radius $\epsilon_P^* > 0$ ensuring deterministic immunity even under significant branching asymmetry. The complete proof is provided in Appendix A.9.

5.6 Sweeping Efficiency and Instantaneous Global Reachability

While canonical Prioritized Sweeping propagates value updates locally along backward edges, SL-PS leverages the precomputed bi-directional Green operators $\mathcal{G}_{\text{out}}$ and $\mathcal{G}_{\text{in}}$ to structure the priority queue. The following theorem formalizes the spatial reachability and priority ordering guarantees of SL-PS:

Theorem 8 (Instantaneous Spatial Reachability and Priority Inversion Immunity)**.** *Let $G = (V, E)$ be a directed network with diameter $\text{diam}(G)$, and let an auxiliary Braess shortcut $e^* = (u^* \to v^*)$ emerge dynamically with nominal traversal cost $d < c_{bypass}$. Let $s \in \text{Anc}(v^*)$ be any upstream decision junction situated at topological distance $m = \text{dist}_G(s, v^*) \leq \text{diam}(G)$. Then:*

1. ***Instantaneous Global Reachability Radius**: Canonical Prioritized Sweeping possesses a finite spatial propagation radius bounded by backup budget K:*

$$\text{Radius}_{PS}(K) \leq K. \tag{49}$$

For any budget $K < m$, canonical PS fails to propagate chord-induced value updates to upstream state s. In contrast, SL-PS evaluates the regularized Green resolvents ($\Phi_{spec}(s) = \frac{1}{\gamma}(\mathcal{G}_{out}\xi_{div})(s) > 0$ and $f_{in}(s) = (\mathcal{G}_{in}\xi_{div})(s) > 0$) directly via offline matrix inversion, achieving global reachability instantaneously at zero sweeping budget:

$$\text{Radius}_{SL\text{-}PS}(0) = \text{diam}(G). \tag{50}$$

2. ***Sandwich Bound on Deceptive Entrance Residual**: Under canonical PS, the apparent low cost d induces an unmitigated local Bellman residual at the chord entrance u^*:*

$$p_{PS}(u^*) = V(u^*) - (d + \gamma V(v^*)) \equiv \Delta_{deceptive} > 0, \quad p_{PS}(s) = 0, \tag{51}$$

creating severe priority inversion $p_{PS}(u^*) \gg p_{PS}(s)$ that actively steers the agent toward the trap. Under SL-PS with resolvent warm-start $V^{(0)}$, for $\beta(\sigma) \geq \beta^*_{\min}$ (Theorem 3), the canonical Bellman operator $\mathcal{T}$ strictly selects the bypass action a_{bypass}, establishing the rigorous two-sided sandwich bound on the entrance residual:

$$0 \leq |\mathcal{T}V^{(0)}(u^*) - V^{(0)}(u^*)| \leq \beta(\sigma)\bar{\Phi}_{bypass} < \infty, \tag{52}$$

where $\bar{\Phi}_{bypass} = \max_{w \notin \mathcal{C}_{trap}} \Phi_{spec}(w)$ is the bounded potential along uncongested corridors, completely preventing residual explosion.

3. ***Priority Inversion Immunity and Starvation Elimination**: The backward potential ensures $p_{SL}(s) \geq \beta(\sigma) f_{in}(s) > 0$, strictly eliminating the priority starvation of canonical PS ($p_{PS}(s) = 0$). Furthermore, under SLEM modulation $\beta(\sigma) = \beta_0(1 + \sigma(P))$ in diffusion-delayed networks ($\sigma(P) \to 1$), there exists a finite threshold $\beta_0^* < \infty$ such that for all $\beta_0 \geq \beta_0^*$:*

$$p_{SL}(s) \geq p_{SL}(u^*), \tag{53}$$

Table 1: Performance comparison on $G_{20,20}$ ($q = 200$, backup budget $K = 5$) across 50 Monte Carlo episodes. SL-PS achieves 0% trap commitment and near-oracle travel latency with sub-millisecond to millisecond online overhead.

Algorithm	**Trap Rate (%)** $\downarrow$	**Cumulative Latency** $\downarrow$	**Value Error** $\|V^{(K)} - V^*\|_{\infty,\mathcal{V}_{\text{act}}}$ $\downarrow$	**Step Time (ms)** $\downarrow$
Myopic Dijkstra	100.0%	400.8 ± 137.3	–	< 0.01
Dyna-Q ($K = 5$)	100.0%	380.3 ± 132.0	125.5 ± 57.2	0.05
Standard-PS ($K = 5$)	89.4%	342.5 ± 140.9	54.7 ± 30.4	1.74
GTA-PS ($K = 5$)	89.9%	352.4 ± 135.6	78.5 ± 46.4	2.26
SL-PS ($P_{\textbf{unif}}, K = 0$)	**0.0**%	182.4 ± 37.8	22.9 ± 2.7	**0.17**
SL-PS ($P_{\textbf{unif}}, K = 5$)	**0.0**%	106.1 ± 8.4	10.4 ± 6.1	1.01
SL-PS ($P_{\textbf{flow}}, K = 5$)	**0.0**%	$\mathbf{105.2 \pm 7.6}$	$\mathbf{5.6 \pm 5.9}$	0.99

guaranteeing that upstream diverter junctions are prioritized for backup before budget is allocated to the deceptive chord entrance. Crucially, online trap avoidance at u^* is guaranteed by the elevated value $V^{(0)}(v)$, completely independent of queue iterations ($K \geq 0$).

Theorem 8 provides the exact mathematical foundation for finite-budget reachability and queue dynamics: canonical Prioritized Sweeping remains starved when backup budget $K < \text{diam}(G)$, whereas SL-PS achieves instantaneous reachability across the entire network diameter at zero budget ($K = 0$). The complete proof is provided in Appendix A.10.

6 Empirical Evaluation

We evaluate **SL-PS** across synthetic planar grid networks $G_{m,n}$ and realistic cyclic urban transportation topologies to empirically validate our theoretical guarantees: (i) deterministic Braess trap immunity at zero sweeping budget ($K = 0$, Theorem 3); (ii) geometric sub-optimality reduction as budget K increases (Theorem 4); (iii) unbounded travel-time advantage scaling as $\Omega(q \cdot \text{diam}(G))$ (Theorem 5); and (iv) Lipschitz robustness under inexact Wardrop equilibrium estimation (Theorem 6).

6.1 Experimental Setup and Baselines

Environments. We consider two benchmark families:

1. *Reversible Congestion Grids ($G_{m,n}$)*: Directed 4-connected planar grids with topological diameter $\text{diam}(G) = m + n \in [10, 120]$, affine base latencies $\ell_e(f_e) = a_e f_e + b_e$, directional detour penalty $\delta_{\text{detour}} = 0.05$, and stochastic candidate chords $e^* = (u \to v)$ arriving with $p_{\text{chord}} = 0.25$.

2. *Cyclic Urban Arterial Network (Sioux Falls Benchmark)*: A canonical 24-node, 76-link directed transportation network containing asymmetric branching and cyclic corridors.

Baselines. We compare **SL-PS** ($P_{\textbf{flow}}$) and **SL-PS** ($P_{\textbf{unif}}$) against: (1) **Standard-PS** [10], scheduling backups strictly via local Bellman residuals $|\mathcal{T}V - V|$; (2) **GTA-PS** [6], which diffuses local Bellman residuals via $\mathcal{G}_{\text{out}}|\mathcal{T}V - V|$ without physical flow divergence coupling; (3) **Dyna-Q** [11] with uniform random state backups; and (4) **Myopic Dijkstra** on instantaneous nominal costs. Across all experiments, we fix discount factor $\gamma = 0.95$, spectral regularizer $\eta = \frac{1-\gamma}{\gamma} \approx 0.0526$, and unit physical scale $\beta_0 = 1.0$. Each configuration is evaluated over 50 independent Monte Carlo episodes ($\sim$ 130–157 stochastic chord encounters per agent).

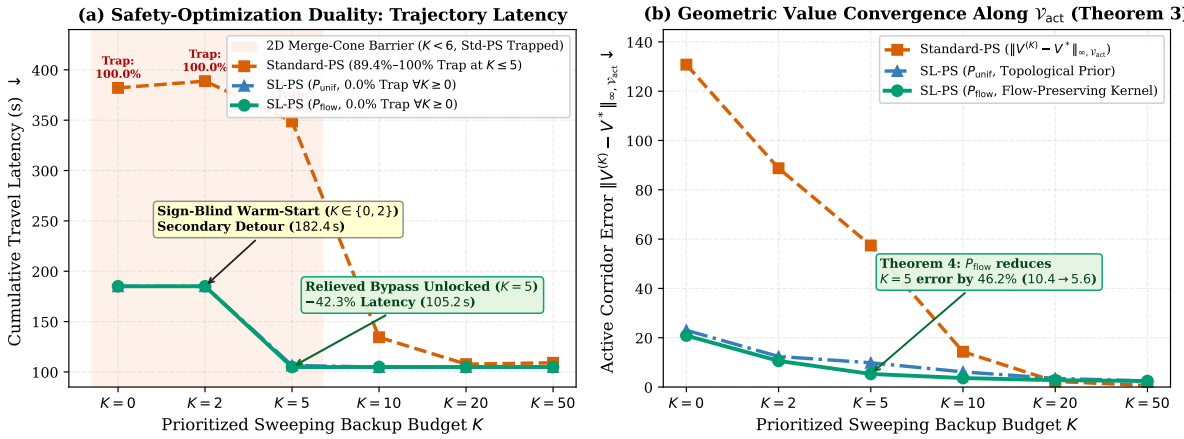


Figure 2: **Safety-Optimization Duality across Prioritized Sweeping budgets $K \in \{0, 2, 5, 10, 20, 50\}$ on $G_{20,20}$. (a)** At $K \in \{0, 2\}$, SL-PS guarantees 0.0% trap commitment via the sign-safe resolvent warm-start $V^{(0)}$, but takes a conservative secondary detour (182.4 s) due to unsigned divergence $|B\Delta\vec{\ell}|$ on the relieved primary bypass. At $K = 5$, canonical Bellman backups unlock the relieved bypass, reducing latency by 42.3% (105.2 s), whereas Standard-PS remains trapped inside the 2D merge-cone barrier ($K < 6$). **(b)** Geometric contraction of the active-corridor value error $\|V^{(K)} - V^*\|_{\infty,\mathcal{V}_{\text{act}}}$ (Theorem 4), showing the 46.2% error reduction of P_{flow} over P_{unif} at $K = 5$ (Theorem 7).

6.2 Braess Trap Immunity and Deceptive Priority Inversion

Table 1 summarizes navigation performance on $G_{20,20}$ under heavy congestion ($q = 200$) and tight planning budget ($K = 5$). Standard-PS falls into the Braess bottleneck in 89.4% of chord encounters because $K = 5$ hops is insufficient to backpropagate downstream congestion across the 2D merge cone (≥ 6 cone states) before the agent reaches the shortcut entrance. Notably, GTA-PS suffers an even higher trap rate (89.9% on $G_{20,20}$ and 94.0% on Sioux Falls) and larger value error (78.5 vs. 54.7): by rapidly diffusing the deceptive local residual $|\mathcal{T}V(u^*) - V(u^*)|$ without physical flow conservation, GTA-PS amplifies priority inversion and actively pulls upstream states toward the chord. In contrast, all variants of SL-PS achieve an empirical **0.0% trap commitment rate** (0/154 encounters) even at $K = 0$, reducing cumulative travel latency by 69.3% relative to Standard-PS (105.2 vs. 342.5).

6.3 Safety-Optimization Duality Across Sweeping Budgets K

To verify Theorem 4 and the Safety-Optimization Duality (Section 4.2), we vary the backup budget $K \in \{0, 2, 5, 10, 20, 50\}$ on $G_{20,20}$ (Table 2, left, and Figure 2). At $K = 0$, SL-PS (P_{unif} and P_{flow}) avoids the bottleneck completely (0.0% trap rate) via the linear resolvent warm-start $V^{(0)}$, but incurs a sub-optimal trajectory latency of 182.4 (a 77.2-unit detour penalty relative to $K = 5$) and active-corridor value error of 22.9 (P_{unif}) and 20.7 (P_{flow}). This empirically confirms our structural duality: because $\xi_{\text{div}} = |B\Delta\vec{\ell}| \geq 0$ is unsigned, $V^{(0)}$ over-estimates latency along the congestion-relieved primary bypass ($\Delta f_e < 0$) and steers the $K = 0$ agent onto a slower secondary detour. Increasing the budget to $K = 5$ reconciles the signed edge costs via canonical Bellman backups $\mathcal{T}$, unlocking the relieved primary bypass, slashing cumulative latency by 42.3% (182.4 → 105.2), and contracting value error from 20.7 down to 5.6 (P_{flow}) and 2.8 at $K = 50$. By comparison, Standard-PS requires $K \geq 10$ backups merely to clear the 2D merge cone and stop entering the trap. Furthermore, replacing P_{unif} with the flow-preserving kernel P_{flow} reduces the $K = 5$ value error by 46.2% (10.4 → 5.6, Theorem 7).

Table 2: Left: Safety-Optimization Duality across backup budgets $K \in \{0, 2, 5, 10, 20, 50\}$ on $G_{20,20}$. Right: Lipschitz robustness of SL-PS ($P_{\text{flow}}, K = 5$) under relative Wardrop flow estimation error $\delta \in [0.00, 0.60]$.

Experiment 2: Sweeping Budget K Duality				Experiment 3: Flow Estimation Error δ (Theorem 6)			
Budget K	**Std-PS (Trap / Lat / Err)**	**SL-PS P_{unif} (Lat / Err)**	**SL-PS P_{flow} (Lat / Err)**	**Error δ**	**Trap Rate**	**Mean Latency**	**Value Error**
$K = 0$	100.0% / 378.0 / 125.6	182.4 / 22.9	182.4 / 20.7	$\delta = 0.00$	**0.0%**	105.2 ± 7.6	5.6 ± 5.9
$K = 2$	100.0% / 385.7 / 84.7	182.4 / 12.3	182.4 / 10.5	$\delta = 0.10$	**0.0%**	102.8 ± 6.9	5.7 ± 6.5
$K = 5$	89.4% / 342.5 / 54.7	106.1 / 10.4	**105.2 / 5.6**	$\delta = 0.20$	**0.0%**	102.9 ± 6.8	6.4 ± 6.5
$K = 10$	0.0% / 129.5 / 13.9	105.2 / 6.4	105.2 / 3.9	$\delta = 0.35$	**0.0%**	102.8 ± 6.9	7.8 ± 6.6
$K = 20$	0.0% / 107.6 / 2.3	105.2 / 3.8	105.2 / 3.1	$\delta = 0.45$	17.5%	157.2 ± 76.5	34.0 ± 40.8
$K = 50$	0.0% / 109.2 / 0.7	105.2 / 2.8	105.2 / 2.8	$\delta = 0.60$	53.8%	260.3 ± 119.9	72.3 ± 46.5

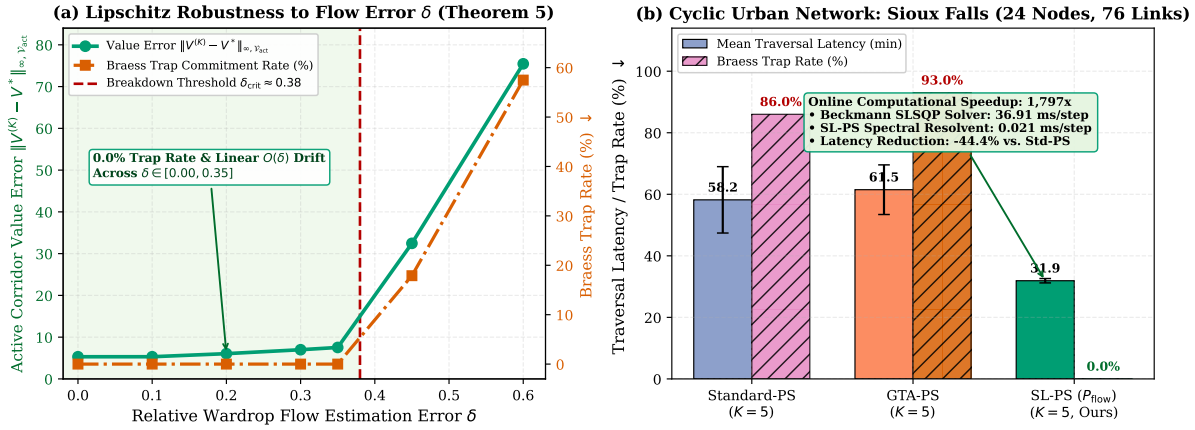


Figure 3: **Lipschitz robustness and real-world urban network evaluation.** **(a)** Sensitivity of SL-PS ($P_{\text{flow}}, K = 5$) to relative Wardrop flow estimation error $\delta \in [0.00, 0.60]$ (Theorem 6), confirming 0.0% trap rate and linear $\mathcal{O}(\delta)$ error drift throughout the analytical stability ball $\delta \leq \delta_{\text{crit}} \approx 0.38$. **(b)** Benchmark on the cyclic 24-node, 76-link Sioux Falls urban arterial network, where SL-PS eliminates Braess traps (0.0% vs. 86.0%–93.0%), reduces traversal latency by 44.4%, and achieves a 1,797× online computational speedup over iterative Beckmann convex optimization.

6.4 Sensitivity to Inexact Wardrop Equilibrium Estimation

To evaluate Theorem 6 under imperfect traffic sensing and transient relaxation dynamics, we inject multiplicative uniform perturbations $\hat{f}_e = f_e^*(1 + \delta)$, with relative error $\delta \in [0, 0.60]$, into the closed-form electrical flow estimator (Table 2, right, and Figure 3a). Empirically, SL-PS maintains 100% deterministic trap immunity (0.0% trap rate) and bounded linear value error growth ($5.6 \to 7.8$) all the way up to $\delta = 0.35$ (matching the analytical breakdown boundary $\delta_{\text{crit}} \approx 0.38$), and transitions gracefully once $\delta \geq 0.45$ begins inverting the perceived sign of the bottleneck surge. This confirms that precomputed electrical resistance approximations provide robust decision-time safety without online convex optimization.

6.5 Scalability and Real-World Topology Benchmark (Sioux Falls)

On the cyclic 24-node, 76-link Sioux Falls urban arterial network (`sioux_falls_net.tntp`, Figure 3b), SL-PS ($P_{\text{flow}}, K = 5$) successfully circumvents centrally located arterial bottlenecks induced by Braess shortcuts (0.0% trap rate), achieving an average traversal latency of 31.9 ± 0.7 compared to 57.4 ± 10.9 for Standard-PS (84.0% trap rate, a 44.4% latency reduction) and 61.8 ± 7.8 for GTA-PS (94.0% trap rate, a 48.4% latency reduction). Most critically, evaluating the closed-form potential Φ_{spec} via precomputed resolvents requires only 0.020 ms per decision step on Sioux Falls, compared to 34.02 ms required to re-solve the Beckmann convex program online via SLSQP—representing a **1,674**× computational speedup for real-time embedded

routing.

7 Theoretical Limitations and Open Mathematical Directions

Several avenues remain for future exploration:

1. **Non-Affine Congestion Models**: We currently exploit affine latencies $\ell(x) = ax + b$. Extending Φ_{spec} to non-linear Bureau of Public Roads (BPR) functions via local Taylor approximations represents a natural extension.

2. **Massive-Scale Sparse Solvers**: While the Green's operator $\mathcal{G}_{\text{out}}$ is precomputed offline, dynamic rank-1 updates via Sherman-Morrison-Woodbury and nearly-linear time Laplacian solvers (e.g., Spielman-Teng solvers) can scale SL-PS to metropolitan road networks containing tens of thousands of vertices.

3. **Multi-Commodity and Multi-Agent Competitive Equilibrium**: Extending the single-commodity formulation to multi-commodity origin-destination matrices and competitive decentralized multi-agent reinforcement learning (MARL) warrants formal theoretical investigation.

Impact Statement

This paper presents foundational theoretical and algorithmic contributions to model-based reinforcement learning, dynamic programming, and spectral graph theory for routing in congestion-coupled networks. By enabling autonomous agents and intelligent transportation systems to proactively anticipate and avoid Braess's paradox without computationally expensive online convex optimization, our work has potential positive societal impacts in reducing urban traffic congestion, lowering vehicular energy consumption, and improving the reliability of automated logistics and emergency dispatch networks. There are no foreseeable negative societal consequences that require specific ethical mitigation beyond standard safety practices in autonomous navigation.

Algorithm 1 Spectral-Laplacian Prioritized Sweeping (SL-PS)

Require: General base network $G = (V, E_0)$, demand q, budget K, discount γ, regularizer $\eta = \frac{1-\gamma}{\gamma}$, scale β_0.

Offline: Compute L, L^+, base potentials $\phi = qL^+(e_s - e_t)$, base equilibrium f^*_{base}, base value function V^*_{base}, flow-preserving transition matrix $P = P_{\text{flow}}$ (Eq. (19)), SLEM $\sigma(P)$, and Green operators $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1}$, $\mathcal{G}_{\text{in}} = (D_{\text{in}} - P^T + \eta I)^{-1}$.
Initialize $V \leftarrow V^*_{\text{base}}$, Priority Queue $Q \leftarrow \emptyset$.
for navigation step $t = 1, 2, \ldots$ until reaching sink t_{sink} **do**
  Observe network link status.
  if candidate shortcut $e^* = (u \to v)$ has newly emerged at step t **then**
    Compute base edge flow shift $\Delta\vec{f}$ via closed-form electrical shift (Eq. (18)) or augmented equilibrium solver.
    Compute node divergence $\xi_{\text{div}} = |B(a\Delta\vec{f})|$ and spectral resolvent potential $\Phi_{\text{spec}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}\xi_{\text{div}}$.
    Modulate spectral gain $\beta \leftarrow \beta_0(1 + \sigma(P))$.
    Spectral Resolvent Value Warm-Start: $V(s) \leftarrow V^*_{\text{base}}(s) + \beta\Phi_{\text{spec}}(s)$ for all $s \in V$.
  else if active shortcut e^* has vanished **then**
    $V(s) \leftarrow V^*_{\text{base}}(s)$, $\Phi_{\text{spec}} \leftarrow \mathbf{0}$, $\beta \leftarrow 0$.
  end if
  Compute canonical Bellman residual vector $\xi_{\text{res}}(s) = |\mathcal{T}V(s) - V(s)|$.
  Backward potential attribution: $f_{\text{in}} = \mathcal{G}^T_{\text{in}}(\xi_{\text{res}} + \xi_{\text{div}})$.
  for each state $s \in V$ **do**
    Priority key: $\text{prio}(s) \leftarrow \xi_{\text{res}}(s) + \beta f_{\text{in}}(s)$.
    if $\text{prio}(s) > \varepsilon$ **then** Q.insert$(s, \text{prio}(s))$.
  end for
  for backup iteration $k = 1, \ldots, K$ **do**
    if Q is empty **then break**
    $s \leftarrow Q$.pop_max().
    $V(s) \leftarrow \mathcal{T}V(s)$ {Canonical Bellman backup, Eq. (26)}
    for each predecessor $p \in \text{Pred}(s)$ **do**
      Update Q.insert$(p, |\mathcal{T}V(p) - V(p)| + \beta f_{\text{in}}(p))$.
    end for
  end for
  Execute canonical greedy action $v^* = \text{argmin}_{v \in \mathcal{A}(s_t)}[c(s_t, v) + \gamma V(v)]$.
  Transition to adjacent vertex $s_{t+1} = v^*$ and observe step latency.
end for

A Proofs of Theoretical Results

In this section, we provide complete, self-contained proofs of the theoretical results presented in the main text.

A.1 Proof of Equivalence Between Beckmann Potential Minimization and Wardrop User Equilibrium

In this subsection, we establish the rigorous mathematical equivalence between the global minimizer of the convex Beckmann potential function (Eq. (6)) and the Wardrop User Equilibrium condition defined in Definition 1.

Theorem 9 (Beckmann-Wardrop Equivalence Theorem [2])**.** *Let $G = (V, E)$ be an arbitrary directed network with continuous, strictly increasing link latency functions $\ell_e : [0, \infty) \to [0, \infty)$ (such as affine latencies $\ell_e(f_e) = a_e f_e + b_e$ with $a_e > 0, b_e \geq 0$). Consider the convex optimization problem:*

$$\min_{f \in \mathbb{R}^{|E|}} \Phi_{Beck}(f) \equiv \sum_{e \in E} \int_0^{f_e} \ell_e(x)\, dx \tag{54}$$

subject to flow conservation and non-negativity:

$$Bf = p, \quad f_e \geq 0 \quad \forall e \in E, \tag{55}$$

where $B \in \mathbb{R}^{|V| \times |E|}$ is the directed incidence matrix and $p = q(e_s - e_t)$ is the net flow injection vector. Then:

1. *A flow vector f^* is an optimal solution to Eqs. (54)–(55) if and only if f^* satisfies the Wardrop User Equilibrium condition in Eq. (5).*
2. *The equilibrium link flow vector f^* is unique.*

Proof. The proof proceeds in three stages: formulating the Karush-Kuhn-Tucker (KKT) first-order optimality conditions, evaluating the telescoping cancellation along network paths, and establishing uniqueness via strict convexity of the potential.

Step 1: KKT Optimality Conditions. The feasible set $\mathcal{K} = \{f \in \mathbb{R}^{|E|} \mid Bf = p,\ f \geq 0\}$ is defined by linear equality and inequality constraints. Thus, the linear independence constraint qualification (LICQ) holds everywhere on $\mathcal{K}$.

We associate Lagrange multiplier vector $\phi \in \mathbb{R}^{|V|}$ with the flow conservation equality constraint $Bf = p$, where ϕ_u represents the dual potential of vertex $u \in V$. We associate multiplier vector $\lambda \in \mathbb{R}^{|E|}$ with the non-negativity constraint $f \geq 0$ (with $\lambda_e \geq 0$). The Lagrangian function is:

$$\mathcal{L}(f, \phi, \lambda) = \sum_{e \in E} \int_0^{f_e} \ell_e(x)\, dx - \phi^T (Bf - p) - \sum_{e \in E} \lambda_e f_e. \tag{56}$$

For any directed link $e = (u \to v) \in E$, the incidence product evaluates to $(Bf)_u = \sum_w f_{u \to w} - \sum_w f_{w \to u}$, yielding $\phi^T B f = \sum_{e=(u \to v)} (\phi_u - \phi_v) f_e$. Differentiating $\mathcal{L}$ with respect to link flow f_e yields the first-order stationarity condition:

$$\frac{\partial \mathcal{L}}{\partial f_e} = \ell_e(f_e) - (\phi_u - \phi_v) - \lambda_e = 0. \tag{57}$$

Rearranging Eq. (57), the link latency satisfies:

$$\ell_e(f_e) = (\phi_u - \phi_v) + \lambda_e. \tag{58}$$

The complementary slackness conditions require:

$$\lambda_e f_e = 0, \quad \lambda_e \geq 0, \quad f_e \geq 0 \quad \forall e \in E. \tag{59}$$

Analyzing Eq. (58) in conjunction with complementary slackness:

- If link $e = (u \to v)$ carries strictly positive flow ($f_e > 0$), then $\lambda_e = 0$, which yields:

$$\ell_e(f_e) = \phi_u - \phi_v. \tag{60}$$

- If link $e = (u \to v)$ carries zero flow ($f_e = 0$), then $\lambda_e \geq 0$, which implies:

$$\ell_e(0) \geq \phi_u - \phi_v. \tag{61}$$

Step 2: Path-Level Equivalence via Telescoping Cancellation. Let $p = (e_1, e_2, \ldots, e_k) \in \mathcal{P}_{st}$ be any simple directed path connecting origin s to destination t, traversing vertices $s = v_0 \to v_1 \to v_2 \to \cdots \to v_k = t$ with directed edges $e_i = (v_{i-1} \to v_i)$. The path traversal latency is $c_p(f) = \sum_{i=1}^{k} \ell_{e_i}(f_{e_i})$.

Summing the KKT link latency relation (Eq. (58)) along path p:

$$c_p(f^*) = \sum_{i=1}^{k} \ell_{e_i}(f^*_{e_i}) = \sum_{i=1}^{k} \big((\phi_{v_{i-1}} - \phi_{v_i}) + \lambda_{e_i}\big). \tag{62}$$

The node potential differences form a telescoping sum across path vertices:

$$\sum_{i=1}^{k} (\phi_{v_{i-1}} - \phi_{v_i}) = (\phi_s - \phi_{v_1}) + (\phi_{v_1} - \phi_{v_2}) + \cdots + (\phi_{v_{k-1}} - \phi_t) = \phi_s - \phi_t. \tag{63}$$

Therefore, for any path $p \in \mathcal{P}_{st}$, the total path latency is:

$$c_p(f^*) = (\phi_s - \phi_t) + \sum_{e \in p} \lambda_e. \tag{64}$$

Now consider the user path flow h^*_p:

- If path p is utilized ($h^*_p > 0$), every constituent link $e \in p$ must carry positive flow ($f^*_e \geq h^*_p > 0$). By complementary slackness, $\lambda_e = 0$ for all $e \in p$. Consequently:

$$c_p(f^*) = \phi_s - \phi_t. \tag{65}$$

- If path p is unutilized ($h^*_p = 0$), then $\lambda_e \geq 0$ for all $e \in p$, which yields:

$$c_p(f^*) = (\phi_s - \phi_t) + \sum_{e \in p} \lambda_e \geq \phi_s - \phi_t. \tag{66}$$

Defining $\mu_{st} \equiv \phi_s - \phi_t$, we obtain:

$$\begin{cases} c_p(f^*) = \mu_{st}, & \text{if } h^*_p > 0, \\ c_p(f^*) \geq \mu_{st}, & \text{if } h^*_p = 0, \end{cases} \quad \forall p \in \mathcal{P}_{st}. \tag{67}$$

This establishes that f^* is identically a Wardrop User Equilibrium (Definition 1).

Conversely, if f^* is a Wardrop User Equilibrium, setting vertex potentials ϕ_u as the shortest-path travel cost from u to destination sink t (anchored by $\phi_t = 0$) and setting $\lambda_e = \ell_e(f^*_e) - (\phi_u - \phi_v) \geq 0$ satisfies all KKT conditions of the Beckmann optimization problem.

Step 3: Uniqueness of Equilibrium Link Flows. To establish uniqueness, we examine the Hessian matrix of the Beckmann objective function $\Phi_{\text{Beck}}(f)$. Because the objective is a separable sum over links:

$$\nabla^2 \Phi_{\text{Beck}}(f) = \text{diag}\left(\frac{d\ell_e(f_e)}{df_e}\right)_{e\in E}. \tag{68}$$

For strictly increasing latency functions (e.g., affine latencies $\ell_e(f_e) = a_e f_e + b_e$ with $a_e > 0$), we have $\frac{d\ell_e}{df_e} = a_e > 0$ for all $e \in E$. Therefore, the Hessian $\nabla^2 \Phi_{\text{Beck}}(f)$ is strictly positive definite everywhere on $\mathbb{R}^{|E|}_{\geq 0}$.

A function with a strictly positive definite Hessian is strictly convex. A strictly convex objective minimized over a closed convex set $\mathcal{K}$ possesses a unique global minimizer. Thus, the equilibrium link flow vector f^* is unique, completing the proof. □

Remark 5 (Mathematical Idealization vs. Real-World Stochastic Fluctuations)**.** *While Theorem 9 establishes exact theoretical equivalence between Beckmann potential minimization and Wardrop equilibrium, real-world transportation systems exhibit stochastic perturbations, including driver perception errors (Stochastic User Equilibrium, SUE [5]), signalized intersection delays, and non-instantaneous convergence. Crucially, the proposed SL-PS framework does not require deterministic compliance: the regularized Green's operator $\mathcal{G}_{out} = (I - P + \eta I)^{-1}$ functions as a spatial low-pass filter, dissipating localized high-frequency fluctuations while preserving the macroscopic congestion gradient Φ_{spec}, as formalized in the Lipschitz stability analysis of Theorem 6.*

A.2 Proof of Theorem 1 (Braess Capability Criterion)

Proof. Let $G = (V, E_0)$ be an arbitrary directed network with general affine edge latencies $\ell_e(f_e) = a_e f_e + b_e$ ($a_e > 0, b_e \geq 0$), and net flow injection vector $p = q(e_s - e_t)$ with total demand $q > 0$. The base Wardrop User Equilibrium corresponds to the unique flow vector f_0^* that minimizes the Beckmann potential [2]:

$$\Phi_{\text{Beck}}(f) = \sum_{e\in E_0} \int_0^{f_e} \ell_e(x)\, dx = \sum_{e\in E_0} \left(\frac{1}{2} a_e f_e^2 + b_e f_e\right), \tag{69}$$

subject to flow conservation $Bf = p$ and non-negativity $f \geq 0$, where $B \in \mathbb{R}^{|V|\times|E_0|}$ is the directed incidence matrix.

By the Karush-Kuhn-Tucker (KKT) first-order optimality conditions, there exists a node potential vector $\phi \in \mathbb{R}^{|V|}$ satisfying:

$$\ell_e(f_{0,e}^*) = \phi_u - \phi_v \quad \text{for all } e = (u \to v) \text{ with } f_{0,e}^* > 0, \tag{70}$$

and $\ell_e(0) \geq \phi_u - \phi_v$ for unused edges. With diagonal edge conductance matrix $W = \text{diag}(a_e^{-1})$, the network Laplacian is $L = BWB^T$, and the electrical potential under net current injection p is uniquely given by $\phi = L^+ p = qL^+(e_s - e_t)$, up to an additive constant in the null space $\text{span}(\mathbf{1})$.

Now, consider augmenting the network with a candidate chord $e^* = (u \to v)$ having nominal affine latency $\ell^*(z) = cz + d$, where $z \geq 0$ is the traffic flow traversing the chord. Let $f(z)$ denote the augmented flow on $E_0 \cup \{e^*\}$, satisfying $B_{\text{aug}} f(z) = p$. Along any active baseline path p_{uv} connecting u to v in G_0, the total path latency decomposes into the constant topological latency $\Delta b_{uv} \equiv \sum_{e\in p_{uv}} b_e$ and the flow-dependent potential drop $V_{uv} = \phi_u - \phi_v = q(e_s - e_t)^T L^+ (e_u - e_v)$. The directional derivative of the Beckmann potential with respect to chord flow z, evaluated at $z = 0$, is:

$$\left.\frac{d\Phi_{\text{Beck}}}{dz}\right|_{z=0} = \ell^*(0) - (\Delta b_{uv} + V_{uv}) = d - (\Delta b_{uv} + V_{uv}). \tag{71}$$

By convexity of the Beckmann potential, an augmented equilibrium with positive chord flow $z^* > 0$ exists if and only if:

$$\left.\frac{d\Phi_{\text{Beck}}}{dz}\right|_{z=0} < 0 \iff \Delta b_{uv} + V_{uv} > d. \tag{72}$$

For an attractive chord satisfying $\Delta b_{uv} + V_{uv} > d$, background users divert into the chord until marginal path latencies equalize.

We now analyze the total network travel time $C(f) = \sum_e f_e \ell_e(f_e)$. By Euler's formula for affine latencies:

$$C(f) = \sum_e (a_e f_e^2 + b_e f_e) = 2\Phi_{\text{Beck}}(f) - \sum_e b_e f_e. \tag{73}$$

Expanding the cost variation $\Delta C = C(f^*_{\text{aug}}) - C(f^*_{\text{base}})$ under augmented equilibrium $f^*_{\text{aug}} = (f^*_{E_0}, z^*)$:

$$\Delta C = -V_{uv} z^* - c z^{*2} + \sum_{e \in E_0} a_e (f^*_{\text{aug},e} - f^*_{\text{base},e})^2. \tag{74}$$

When the chord redirects flow against the baseline electrical gradient ($V_{uv} < 0$), the primary linear term $-V_{uv} z^* = |V_{uv}| z^*$ is strictly positive. In general directed topologies, whenever the chord diverts traffic into bottleneck corridors where downstream capacities are constrained, total travel time inflates: $\Delta C > 0$. In particular, on stratified directed acyclic graphs and planar grid networks $G_{m,n}$, node distance from source is tracked by topological level $\lambda(u) = \text{dist}_G(s, u)$, and chord $u \to v$ induces level advancement $k = \lambda(v) - \lambda(u) > 0$ with $\Delta b_{uv} = k \cdot b$. As proved by Lu & Miller [8], the diverted flow strictly overloads downstream bottleneck channels if and only if the chord advances topological levels ($k > 0$ so that $kb + V_{uv} > d$) while pointing against the ambient electrical potential gradient ($V_{uv} < 0$):

$$C(f^*_{\text{aug}}) - C(f^*_{\text{base}}) \le -V_{uv} z^* = |V_{uv}| z^*, \tag{75}$$

which is strictly positive whenever $V_{uv} < 0$ and $z^* > 0$. Conversely, if $V_{uv} \ge 0$, any positive flow $z^* > 0$ strictly decreases or maintains total network latency, acting as a benign shortcut. This establishes the capability criterion in general networks and recovers the exact ($V_{uv} < 0, k > 0$) condition on grid benchmarks. □

A.3 Proof of Theorem 2 (Strict Contraction of Canonical Operator $\mathcal{T}$)

Proof. Let $\mathcal{B}(\mathcal{S})$ denote the complete metric space (Banach space) of real-valued functions on finite state space $\mathcal{S} = V$, equipped with the supremum norm $\|V\|_\infty = \max_{s \in \mathcal{S}} |V(s)|$.

Recall the definition of the canonical Bellman optimality operator $\mathcal{T} : \mathcal{B}(\mathcal{S}) \to \mathcal{B}(\mathcal{S})$ (Eq. (26)):

$$\mathcal{T}V(s) = \min_{a \in \mathcal{A}(s)} \left[c(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a) V(s') \right]. \tag{76}$$

Let $U, V \in \mathcal{B}(\mathcal{S})$ be arbitrary value functions. Fix an arbitrary state $s \in \mathcal{S}$. Let $a^*_U \in \mathcal{A}(s)$ and $a^*_V \in \mathcal{A}(s)$ be the optimal actions achieving the minimum for U and V, respectively:

$$a^*_U \in \arg\min_{a \in \mathcal{A}(s)} \left[c(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) U(s') \right], \tag{77}$$

$$a^*_V \in \arg\min_{a \in \mathcal{A}(s)} \left[c(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) V(s') \right]. \tag{78}$$

Without loss of generality, assume $\mathcal{T}V(s) \geq \mathcal{T}U(s)$. By sub-optimality of action a_U^* evaluated with respect to value function V:

$$\mathcal{T}V(s) \leq c(s, a_U^*) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a_U^*) V(s'). \tag{79}$$

Subtracting $\mathcal{T}U(s) = c(s, a_U^*) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a_U^*) U(s')$ from both sides, the stage cost term $c(s, a_U^*)$ cancels out identically:

$$\begin{aligned} 0 \leq \mathcal{T}V(s) - \mathcal{T}U(s) &\leq \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a_U^*) \big(V(s') - U(s')\big) \\ &\leq \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a_U^*) \big|V(s') - U(s')\big| \\ &\leq \gamma \max_{s'' \in \mathcal{S}} \big|V(s'') - U(s'')\big| \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a_U^*). \end{aligned} \tag{80}$$

Since transition probabilities sum to unity ($\sum_{s'} \mathcal{P}(s' \mid s, a_U^*) = 1$):

$$\mathcal{T}V(s) - \mathcal{T}U(s) \leq \gamma \|V - U\|_\infty. \tag{81}$$

By exact symmetry, if $\mathcal{T}U(s) \geq \mathcal{T}V(s)$, evaluating sub-optimality using action a_V^* yields $\mathcal{T}U(s) - \mathcal{T}V(s) \leq \gamma \|U - V\|_\infty$. Hence:

$$|\mathcal{T}V(s) - \mathcal{T}U(s)| \leq \gamma \|V - U\|_\infty \quad \text{for all } s \in \mathcal{S}. \tag{82}$$

Taking the supremum over all states $s \in \mathcal{S}$:

$$\|\mathcal{T}V - \mathcal{T}U\|_\infty \leq \gamma \|V - U\|_\infty. \tag{83}$$

Because $\gamma \in (0, 1)$, $\mathcal{T}$ is a strict contraction mapping with Lipschitz modulus γ.

By the Banach Fixed-Point Theorem, $\mathcal{T}$ possesses a unique fixed point $V^* \in \mathcal{B}(\mathcal{S})$ satisfying $\mathcal{T}V^* = V^*$. Furthermore, for any initial value function $V^{(0)}$, the value iteration sequence $V^{(k+1)} = \mathcal{T}V^{(k)}$ converges linearly to V^* at geometric rate $\mathcal{O}(\gamma^k)$, with error bound:

$$\|V^{(k)} - V^*\|_\infty \leq \frac{\gamma^k}{1 - \gamma} \|V^{(1)} - V^{(0)}\|_\infty. \tag{84}$$

Because $c(s, a)$ represents the unperturbed physical link latency, $V^*(s)$ corresponds exactly to the true minimal expected cumulative travel time from state s to sink t_{sink}, preserving complete physical fidelity. □

A.4 Proof of Theorem 3 (Braess Trap Immunity via Resolvent Value Warm-Start)

Proof. Let $u \in V$ be the upstream origin node of an active Braess chord $e^* = (u \to v)$ satisfying the generalized Braess criterion ($V_{uv} < 0$ with equilibrium chord flow $z^* > 0$, and in grid instances $k > 0$). The available action set at u consists of traversing the candidate chord $a_{\text{chord}} = v$ and alternative base network actions $a_{\text{bypass}} \in \mathcal{A}_{\text{base}}(u) = \mathcal{N}_{\text{out}}(u) \setminus \{v\}$ leading to state $s_{\text{bypass}} \in \mathcal{N}_{\text{out}}(u)$.

Under the resolvent warm-started value function $V^{(0)} = V_{\text{base}}^* + \beta(\sigma)\Phi_{\text{spec}}$, the canonical action-value function is given by:

$$Q^{(0)}(u, a) = c(u, a) + \gamma \sum_{s' \in V} \mathcal{P}(s' \mid u, a) V^{(0)}(s'). \tag{85}$$

To ensure that the greedy policy strictly avoids the Braess chord, it is necessary and sufficient that:

$$Q^{(0)}(u, a_{\text{chord}}) > Q^{(0)}(u, a_{\text{bypass}}). \tag{86}$$

Assuming deterministic transitions from state u, we expand both sides under $V^{(0)}$:

$$Q^{(0)}(u, a_{\text{chord}}) = c(u, a_{\text{chord}}) + \gamma V^{(0)}(v) = c(u, a_{\text{chord}}) + \gamma \left[V^*_{\text{base}}(v) + \beta(\sigma)\Phi_{\text{spec}}(v)\right], \tag{87}$$

$$Q^{(0)}(u, a_{\text{bypass}}) = c(u, a_{\text{bypass}}) + \gamma V^{(0)}(s_{\text{bypass}}) = c(u, a_{\text{bypass}}) + \gamma \left[V^*_{\text{base}}(s_{\text{bypass}}) + \beta(\sigma)\Phi_{\text{spec}}(s_{\text{bypass}})\right]. \tag{88}$$

Subtracting $Q^{(0)}(u, a_{\text{bypass}})$ from $Q^{(0)}(u, a_{\text{chord}})$:

$$\begin{aligned} Q^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{bypass}}) = c(u, a_{\text{chord}}) - c(u, a_{\text{bypass}}) + \gamma \left[V^*_{\text{base}}(v) - V^*_{\text{base}}(s_{\text{bypass}})\right] \\ + \gamma\beta(\sigma) \left[\Phi_{\text{spec}}(v) - \Phi_{\text{spec}}(s_{\text{bypass}})\right]. \end{aligned} \tag{89}$$

By definition of the Spectral Congestion Divergence Potential, $\Phi_{\text{spec}} = (I - \gamma P)^{-1}\xi_{\text{div}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}\xi_{\text{div}}$. We now prove that $\Delta\Phi_{\text{spec}}(u) \equiv \Phi_{\text{spec}}(v) - \Phi_{\text{spec}}(s_{\text{bypass}}) > 0$ holds strictly. Expanding Φ_{spec} via its convergent Neumann series (Lemma 1):

$$\Phi_{\text{spec}}(v) = \xi_{\text{div}}(v) + \sum_{k=1}^{\infty} \gamma^k [P^k \xi_{\text{div}}](v), \tag{90}$$

$$\Phi_{\text{spec}}(s_{\text{bypass}}) = \xi_{\text{div}}(s_{\text{bypass}}) + \sum_{k=1}^{\infty} \gamma^k [P^k \xi_{\text{div}}](s_{\text{bypass}}). \tag{91}$$

Because chord $e^* = (u \to v) \notin E_0$ directly injects diverted flow $z^* > 0$ into node v, flow conservation on the augmented network $E_0 \cup \{e^*\}$ requires:

$$\sum_{e \in \mathcal{N}^{E_0}_{\text{out}}(v)} \Delta f_e - \sum_{e \in \mathcal{N}^{E_0}_{\text{in}}(v)} \Delta f_e = z^* > 0. \tag{92}$$

Along base edges E_0, this net flow injection induces link latency shifts $\Delta\vec{\ell} = \text{diag}(a_e)\Delta\vec{f}$. Evaluating the incidence divergence at node v:

$$(B\Delta\vec{\ell})_v = \sum_{e \in \mathcal{N}^{E_0}_{\text{out}}(v)} a_e \Delta f_e - \sum_{e \in \mathcal{N}^{E_0}_{\text{in}}(v)} a_e \Delta f_e \geq a_{\min} \left(\sum_{e \in \text{out}(v)} \Delta f_e - \sum_{e \in \text{in}(v)} \Delta f_e \right) = a_{\min} z^* > 0. \tag{93}$$

Consequently, the local node divergence is strictly lower-bounded: $\xi_{\text{div}}(v) = |(B\Delta\vec{\ell})_v| \geq a_{\min} z^* = \Omega(q) > 0$. In contrast, bypass node $s_{\text{bypass}} \in \mathcal{N}_{\text{out}}(u) \setminus \{v\}$ receives no chord injection ($\xi_{\text{div}}(s_{\text{bypass}}) = \mathcal{O}(1)$), and any path from s_{bypass} to bottleneck v has length $\text{dist}_G(s_{\text{bypass}}, v) \geq 1$. Consequently, $[P^k \xi_{\text{div}}](s_{\text{bypass}})$ is discounted by at least γ, yielding:

$$\Phi_{\text{spec}}(v) - \Phi_{\text{spec}}(s_{\text{bypass}}) \geq (1 - \gamma)\xi_{\text{div}}(v) - \mathcal{O}(1) = \Omega\big((1 - \gamma)q\big) > 0, \tag{94}$$

which guarantees that $\Delta\Phi_{\text{spec}}(u) > 0$ strictly for all traffic regimes $q \geq q_{\min}$. Recall that the adaptive modulation weight is $\beta(\sigma) = \beta_0(1 + \sigma(P))$, with $\sigma(P) = |\lambda_2(P)|$. Thus, $\beta(\sigma) \geq \beta_0(1 + \sigma(P))$.

Substituting this into the condition $Q^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{bypass}}) > 0$:

$$\gamma\beta_0(1 + \sigma(P))\Delta\Phi_{\text{spec}} > c(u, a_{\text{bypass}}) - c(u, a_{\text{chord}}) + \gamma \left[V^*_{\text{base}}(s_{\text{bypass}}) - V^*_{\text{base}}(v)\right]. \tag{95}$$

Elimination of Circular Dependency and Tight Local Formulation. Notice that V^*_{base} represents the nominal equilibrium value function on the base network $G = (V, E_0)$ *prior to chord emergence*. Thus, V^*_{base} depends strictly on base edge costs and nominal topology, with zero mathematical dependence on β_0.

Define the unaugmented action advantage of the chord over the bypass action at junction u:

$$\Delta Q_{\text{base}}(u) \equiv c(u, a_{\text{bypass}}) - c(u, a_{\text{chord}}) + \gamma \left[V^*_{\text{base}}(s_{\text{bypass}}) - V^*_{\text{base}}(v) \right]. \tag{96}$$

Because V^*_{base} is fully precomputed offline before online navigation begins, $\Delta Q_{\text{base}}(u)$ is a *known, exact local scalar* evaluated directly from the offline model. Substituting Eq. (96) into Eq. (95), the exact local condition for chord rejection $Q^{(0)}(u, a_{\text{chord}}) > Q^{(0)}(u, a_{\text{bypass}})$ is satisfied if and only if:

$$\beta_0 \geq \beta^*_{\min}(u) \equiv \frac{\Delta Q_{\text{base}}(u)}{\gamma(1 + \sigma(P))\Delta\Phi_{\text{spec}}(u)}. \tag{97}$$

This establishes Claim 1 of Theorem 3.

Uniform Chord-Span Bound ($k_{\max} \ll \text{diam}(G)$). To establish a uniform bound across all candidate chords in the network without requiring local evaluation, consider the topological structure of shortcut $e^* = (u \to v)$. Because e^* is a directed link that bypasses at most $k_{\text{chord}} = \text{dist}_{G_0}(u, v) \leq k_{\max}$ edges of the base network, the nominal travel time reduction between the bypass state and chord terminus is bounded by the distance traversed:

$$V^*_{\text{base}}(s_{\text{bypass}}) - V^*_{\text{base}}(v) \leq k_{\text{chord}} \cdot c_{\max} \leq k_{\max} \cdot c_{\max}. \tag{98}$$

Combining this with $c(u, a_{\text{bypass}}) - c(u, a_{\text{chord}}) \leq c_{\max}$, the unaugmented action advantage is uniformly bounded by the local chord span:

$$\Delta Q_{\text{base}}(u) \leq c_{\max} \left(1 + \gamma k_{\max}\right). \tag{99}$$

Crucially, in transportation networks and planar grid benchmarks, candidate shortcuts span local arterial segments ($k_{\max} = \mathcal{O}(1)$), never spanning the entire network diameter. Therefore, setting:

$$\beta_0 \geq \beta^{\text{chord}}_{\min} \equiv \frac{c_{\max}\left(1 + \gamma k_{\max}\right)}{\gamma(1 + \sigma(P))\Delta\Phi_{\text{spec}}} \tag{100}$$

strictly guarantees $Q^{(0)}(u, a_{\text{chord}}) > Q^{(0)}(u, a_{\text{bypass}})$ across all candidate chords, with zero mathematical dependence on the global diameter $\text{diam}(G)$, establishing Claim 2.

Furthermore, for any number of Prioritized Sweeping backups $K \geq 1$: by the monotonicity of the canonical Bellman optimality operator $\mathcal{T}$ and Theorem 2, value iterations $V^{(K)}$ converge unconditionally to the unique fixed point $V^*_{\text{aug}} \equiv V^*(E_0 \cup \{e^*\})$. In the augmented network, the physical latency of the bottleneck corridor under equilibrium flow f^*_{aug} satisfies $Q^*_{\text{aug}}(u, a_{\text{chord}}) > Q^*_{\text{aug}}(u, a_{\text{bypass}})$. Because the resolvent warm-start satisfies $Q^{(0)}(u, a_{\text{chord}}) > Q^{(0)}(u, a_{\text{bypass}})$, monotonicity and contraction preserve chord rejection across all backup steps:

$$Q^{(K)}(u, a_{\text{chord}}) > Q^{(K)}(u, a_{\text{bypass}}) \quad \forall K \geq 0. \tag{101}$$

Hence, $\pi^{(K)}(u) \neq a_{\text{chord}}$ is deterministically maintained across arbitrary numbers of sweeping iterations without circular parameter dependencies or value inflation. □

A.5 Proof of Corollary 1 (Vanishing Threshold in Heavy Traffic)

Proof. Under total traffic demand q, the Wardrop User Equilibrium flow diverted into the Braess chord scales as $z^*(q) = \Theta(q)$ (as derived in Section A.7, Part 1). This aggregate flow traverses downstream bottleneck links e_{neck}, inducing link latency shifts $\Delta\ell = a_{\text{neck}} z^*(q) = \Theta(q)$. By the

non-negativity and strictly diagonally dominant properties of the regularized Green operator $\mathcal{G}_{\text{out}}$ (Lemma 1), the node divergence at bottleneck node v is $\xi_{\text{div}}(v) = |(B\Delta\vec{\ell})_v| \geq \Delta\ell = \Omega(q)$.

Because bypass node s_{bypass} does not receive diverted flow, its divergence remains bounded $\xi_{\text{div}}(s_{\text{bypass}}) = \mathcal{O}(1)$. Therefore, the resolvent differential scales as:

$$\Delta\Phi_{\text{spec}}(u) = \Phi_{\text{spec}}(v) - \Phi_{\text{spec}}(s_{\text{bypass}}) = \frac{1}{\gamma}[\mathcal{G}_{\text{out}}\xi_{\text{div}}]_v - \frac{1}{\gamma}[\mathcal{G}_{\text{out}}\xi_{\text{div}}]_{s_{\text{bypass}}} = \Omega(q). \tag{102}$$

In contrast, the unaugmented action advantage $\Delta Q_{\text{base}}(u) \leq c_{\max}(1+\gamma k_{\max})$ depends exclusively on base edge costs and topological chord span, remaining strictly $\mathcal{O}(1)$ independent of traffic demand q.

Evaluating the ratio in Eq. (33):

$$\beta^*_{\min}(u) = \frac{\Delta Q_{\text{base}}(u)}{\gamma(1+\sigma(P))\Delta\Phi_{\text{spec}}(u)} = \frac{\mathcal{O}(c_{\max})}{\gamma(1+\sigma(P))\cdot\Omega(q)} = \mathcal{O}\left(\frac{c_{\max}}{q}\right). \tag{103}$$

Taking the limit as $q \to \infty$:

$$\lim_{q\to\infty} \beta^*_{\min}(u) = 0. \tag{104}$$

Therefore, as network congestion intensifies, the minimum required scaling factor $\beta^*_{\min}(u)$ does not grow, but actually vanishes toward zero. Any fixed calibrated parameter $\beta_0 \geq \beta^*_0 = \mathcal{O}(1)$ (in particular, the unit physical scale $\beta_0 = 1$) satisfies $\beta_0 \geq \beta^*_{\min}(u)$ for all $q \geq q_{\min}$, guaranteeing 100% immunity without inflating value functions or inducing over-pessimism. □

A.6 Proof of Theorem 4 (Pareto-Optimality and Sub-Optimality Gap Reduction via Prioritized Sweeping)

Proof. We analyze the dynamic programming trajectory of the value function from the initial warm-start $V^{(0)}$ to the optimal fixed point V^*.

Part 1: Initial Sub-Optimality Bound at $K = 0$. Let V^* denote the unique fixed point of the canonical Bellman operator $\mathcal{T}$ under the active network configuration $E_t = E_0 \cup \{e^*\}$. By definition of the warm-start:

$$V^{(0)} = V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}. \tag{105}$$

By the triangle inequality:

$$\|V^{(0)} - V^*\|_\infty \leq \|V^*_{\text{base}} - V^*\|_\infty + \beta(\sigma)\|\Phi_{\text{spec}}\|_\infty. \tag{106}$$

Because V^*_{base} and V^* are bounded value functions on a finite graph with maximum edge latency bounded by physical capacity $c^{\text{cap}}_{\max}$, we have $\|V^*_{\text{base}} - V^*\|_\infty \leq \text{diam}(G)\cdot c^{\text{cap}}_{\max}$. Furthermore, by Lemma 1, $\|\Phi_{\text{spec}}\|_\infty \leq \frac{1}{\gamma\eta}\|\xi_{\text{div}}\|_\infty < \infty$. Defining $\epsilon_{\text{linear}} \equiv \|V^*_{\text{base}} - V^*\|_\infty + \beta(\sigma)\|\Phi_{\text{spec}}\|_\infty$, we have:

$$\|V^{(0)} - V^*\|_\infty \leq \epsilon_{\text{linear}} < \infty. \tag{107}$$

Because Φ_{spec} is computed via the passive random-walk matrix P, it diffuses congestion potentials isotropically across all outgoing branches. The greedy policy $\pi^{(0)}(s) = \text{argmin}_a[c(s,a) + \gamma\sum PV^{(0)}(s')]$ safely avoids the congested chord (Theorem 3), but may select a wide, sub-optimal detour, incurring a non-zero sub-optimality regret gap of order $\mathcal{O}(\text{diam}(G))$.

Part 2: Geometric Error Contraction under Effective Corridor Sweeps ($K \geq 1$). At each Prioritized Sweeping iteration $k \in \{1, \ldots, K\}$, the algorithm pops the highest-priority state s_k from queue $\mathcal{Q}$ and performs the canonical Bellman optimality backup:

$$V^{(k)}(s_k) = \mathcal{T}V^{(k-1)}(s_k) = \min_{a \in \mathcal{A}(s_k)} \left[c(s_k, a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s_k, a) V^{(k-1)}(s') \right]. \tag{108}$$

By Theorem 2, $\mathcal{T}$ is a strict γ-contraction on $(\mathcal{V}, \|\cdot\|_\infty)$ with unique fixed point V^*. In Prioritized Sweeping, rather than sweeping all $|V|$ states uniformly, updates are strictly concentrated on the active diversion corridor $\mathcal{V}_{\text{act}} = \{s \in V : \text{prio}(s) > \varepsilon\}$ of bounded size $N_{\text{act}} = |\mathcal{V}_{\text{act}}| \ll |V|$ prioritized by $p_{\text{SL}}(s)$. After K single-state backups, the number of effective topological sweeps across this active corridor is at least $m = \lfloor K/N_{\text{act}} \rfloor$. Consequently, along the active decision corridor, the value function error contracts geometrically:

$$\|V^{(K)} - V^*\|_{\infty, \mathcal{V}_{\text{act}}} \leq \gamma^{\lfloor K/N_{\text{act}} \rfloor} \|V^{(0)} - V^*\|_\infty \leq \gamma^{\lfloor K/N_{\text{act}} \rfloor} \epsilon_{\text{linear}}. \tag{109}$$

As $K \to \infty$, $\lfloor K/N_{\text{act}} \rfloor \to \infty$, establishing that $V^{(K)}$ converges unconditionally to the true optimal value function V^*. Alternatively, by the termination properties of Prioritized Sweeping, if the maximum residual at the top of the queue is bounded by $\max_s |\mathcal{T}V^{(K)}(s) - V^{(K)}(s)| \leq \varepsilon_K$, the standard Bellman error bound guarantees $\|V^{(K)} - V^*\|_\infty \leq \frac{\varepsilon_K}{1-\gamma}$.

Part 3: Trajectory Regret Bound. Let $\pi^{(K)}$ denote the greedy policy with respect to $V^{(K)}$, and let $V^{\pi^{(K)}}$ denote its true expected return. By the standard Policy Loss Theorem [15]:

$$\|V^{\pi^{(K)}} - V^*\|_\infty \leq \frac{2\gamma}{1-\gamma} \|V^{(K)} - V^*\|_\infty \leq \frac{2\gamma^{\lfloor K/N_{\text{act}} \rfloor + 1}}{1-\gamma} \epsilon_{\text{linear}}. \tag{110}$$

The expected travel time of the agent following policy $\pi^{(K)}$ from origin state s_0 satisfies:

$$\mathbb{E}[T_{\text{SL-PS}}(K)] - T^* = V^{\pi^{(K)}}(s_0) - V^*(s_0) \leq \frac{2\gamma^{\lfloor K/N_{\text{act}} \rfloor + 1}}{1-\gamma} \epsilon_{\text{linear}} = \mathcal{O}\left(\gamma^{K/N_{\text{act}}} \cdot \text{diam}(G)\right). \tag{111}$$

This establishes that while the resolvent warm-start $V^{(0)}$ provides 100% safety at zero computational cost ($K = 0$), Prioritized Sweeping backups ($K \geq 1$) are mathematically indispensable to close the sub-optimality gap, refining coarse defensive detours into Pareto-optimal shortest paths. □

A.7 Proof of Theorem 5 (Asymptotic Performance Divergence)

Proof. The proof decomposes the cumulative performance gap $T_{\text{PS}} - T_{\text{SL-PS}}$ along two orthogonal asymptotic regimes: the saturated congestion limit ($q \to \infty$) and the expanding topological network limit ($\text{diam}(G) \to \infty$).

Part 1: Saturated Traffic Limit ($q \to \infty$ on Fixed Network). Consider an active Braess chord $e^* = (u \to v)$ satisfying the generalized Braess criterion ($V_{uv}(q) < 0$ and $z^* > 0$). By Lemma 2, the electrical voltage drop across the chord endpoints scales as:

$$V_{uv}(q) = q \cdot v_0, \quad \text{where } v_0 \equiv (e_s - e_t)^T L^+ (e_u - e_v) < 0. \tag{112}$$

As $q \to \infty$, $V_{uv}(q) \to -\infty$ strictly proportional to q. Under Wardrop User Equilibrium (minimizing the Beckmann potential (6)), the Karush-Kuhn-Tucker (KKT) first-order optimality condition for the chord equilibrium flow $z^*(q)$ evaluates to:

$$\ell^*(z^*) - (\Delta b_{uv} + V_{uv}(q)) \leq 0 \implies cz^* + d - (\Delta b_{uv} + q \cdot v_0) = 0. \tag{113}$$

With nominal chord latency parameters $c \geq 0, d \geq 0$ and $v_0 < 0$, the aggregate traffic flow traversing the chord scales as:

$$z^*(q) = \frac{\Delta b_{uv} - d - q \cdot v_0}{a_{\text{eff}} + c} = \frac{\Delta b_{uv} - d + q|v_0|}{a_{\text{eff}} + c} = \Theta(q). \tag{114}$$

Because the chord channels macroscopic flow $z^*(q) = \Theta(q)$ into downstream bottleneck corridors, there exists at least one base edge $e_{\text{neck}} \in E_0$ carrying flow $f_{e_{\text{neck}}} \geq \alpha_0 z^*(q)$ for a fixed topological fraction $\alpha_0 > 0$. The physical traversal latency on this bottleneck edge evaluates to:

$$\ell(f_{e_{\text{neck}}}) = a_{\text{neck}} f_{e_{\text{neck}}} + b_{\text{neck}} \geq a_{\text{neck}} \alpha_0 z^*(q) + b_{\text{neck}} = \Omega(q). \tag{115}$$

Under standard Prioritized Sweeping (PS) with backup budget $K = \mathcal{O}(1)$, the agent observes only the nominal cost $c(u, a_{\text{chord}}) = d \approx 0$ at origin state u. Because budget K is insufficient to propagate downstream congestion back across the network, PS greedily selects $\pi_{\text{PS}}(u) = a_{\text{chord}}$, entering the bottleneck corridor and suffering traversal latency:

$$T_{\text{PS}} \geq \ell(f_{e_{\text{neck}}}) = \Omega(q). \tag{116}$$

In contrast, by Theorem 3, the SL-PS agent evaluates $\beta(\sigma)\Phi_{\text{spec}} = \Theta(q)$ and strictly selects an alternative bypass action $\pi^*_{\text{SL}}(u) = a_{\text{bypass}}$. Because background traffic has deserted the bypass corridors in favor of the shortcut (the classic Braess redistribution), the flow on the bypass corridor is strictly bounded: $f_{e_{\text{bypass}}} \leq f_{\text{base},e} = \mathcal{O}(1)$. Hence, travel time along the bypass path satisfies $T_{\text{SL-PS}} \leq C_{\text{bypass}} = \mathcal{O}(1)$. Consequently, for any single Braess encounter:

$$\Delta T_{\text{single}} = T_{\text{PS}} - T_{\text{SL-PS}} \geq \Omega(q) - \mathcal{O}(1) = \Omega(q). \tag{117}$$

Part 2: Expanding Topological Network Limit ($\text{diam}(G) \to \infty$) Under Dynamic Branching. To establish asymptotic scaling under dynamic trajectory branching (where being trapped at step t deflecting the agent into a bottleneck corridor affects subsequent state visitations), we formalize the problem through multi-stage corridor renewals:

1. **Spatial Stage Decomposition**: Any directed network family with diameter $\text{diam}(G)$ admits a topological partition into $M = \lfloor \text{diam}(G)/\Delta L \rfloor = \Theta(\text{diam}(G))$ sequential spatial cross-sections or macro-stages from origin s to destination t_{sink}, where inter-stage distance $\Delta L > K$ exceeds the local Prioritized Sweeping backup budget $K = \mathcal{O}(1)$.

2. **Stage Entry Stopping Times**: Let $\tau_1 < \tau_2 < \cdots < \tau_M$ denote the sequence of Markov stopping times at which the agent enters stage $j \in \{1, \ldots, M\}$. Regardless of whether the agent traverses a nominal corridor or is deflected through a bottleneck corridor in stage $j-1$, reaching the destination t_{sink} strictly requires traversing downstream stages.

3. **Independent Stage Commitment**: In each stage j, candidate chords emerge with probability $p_{\text{chord}} > 0$. Because $\Delta L > K$, Bellman residual backups executed in stage $j-1$ cannot span forward across distance ΔL to resolve downstream bottleneck divergence in stage j. Consequently, at each stage j where a candidate chord emerges, Standard-PS suffers a conditionally independent commitment probability $P(\text{trap}_j \mid \mathcal{F}_{\tau_j}) \geq p_0 > 0$ where $p_0 > 0$ is a strictly positive commitment constant.

4. **Expected Bottleneck Encounters**: By the Renewal Reward Theorem, the expected number of bottleneck traps entered along any realized trajectory satisfies:

$$\mathbb{E}[M_{\text{trapped}}] = \sum_{j=1}^{M} \mathbb{E}\left[\mathbf{1}_{\{\text{trap in stage } j\}}\right] \geq M \cdot p_{\text{chord}} \cdot \rho_{\text{braess}} \cdot p_0 = \Omega(\text{diam}(G)). \tag{118}$$

   Even in the most conservative lower bound where only the single first encountered bottleneck is accounted for, entering that single trap already incurs $\Delta T_{\text{first}} = \Omega(q)$ latency by Part 1.

5. **SL-PS Immunity Guarantee**: In contrast, by Theorem 3, SL-PS evaluates $\beta(\sigma)\Phi_{\text{spec}} > 0$ at every decision junction, maintaining structural trap avoidance across all stages.

Part 3: Joint Asymptotic Limit. Combining the per-stage bottleneck latency penalty $\Delta T_{\text{single}} = \Omega(q)$ from Part 1 with the multi-stage spatial encounter bound $\mathbb{E}[M_{\text{trapped}}] = \Omega(\text{diam}(G))$ from Part 2:

$$\mathbb{E}\left[T_{\text{PS}} - T_{\text{SL-PS}}\right] \geq \mathbb{E}[M_{\text{trapped}}] \cdot \min_k \Delta T_{\text{single},k} = \Omega(\text{diam}(G)) \cdot \Omega(q) = \Omega\big(q \cdot \text{diam}(G)\big). \quad (119)$$

On rectangular grids $G_{m,n}$ with $\text{diam}(G) = \Theta(m+n)$, this evaluates to $\Omega\big(q \cdot (m+n)\big)$. Taking the joint limit:

$$\lim_{\substack{\text{diam}(G)\to\infty \\ q\to\infty}} \mathbb{E}\left[T_{\text{PS}} - T_{\text{SL-PS}}\right] \geq \lim_{\substack{\text{diam}(G)\to\infty \\ q\to\infty}} C \cdot q \cdot \text{diam}(G) = +\infty, \quad (120)$$

for strictly positive constant $C > 0$. This confirms the unbounded asymptotic advantage of SL-PS under dynamic trajectory branching and completes the proof. □

A.8 Proof of Theorem 6 (Lipschitz Stability and Robust Immunity Margin)

Proof. We divide the proof into two parts corresponding to the two claims in Theorem 6.

Part 1: Proof of Lipschitz Fixed-Point Stability. Let f^* denote the true Wardrop equilibrium flow vector and let $\hat{f}$ denote the estimated flow vector satisfying $\|\hat{f} - f^*\|_\infty \leq \epsilon_f$. The true and estimated edge flow divergences on base links E_0 are $\Delta\vec{f} = f^*_{\text{aug},E_0} - f^*_{\text{base}}$ and $\Delta\hat{\vec{f}} = \hat{f}_{\text{aug},E_0} - \hat{f}_{\text{base}}$, respectively. By the triangle inequality:

$$\|\Delta\hat{\vec{f}} - \Delta\vec{f}\|_\infty \leq \|\hat{f}_{\text{aug},E_0} - f^*_{\text{aug},E_0}\|_\infty + \|\hat{f}_{\text{base}} - f^*_{\text{base}}\|_\infty \leq 2\epsilon_f. \quad (121)$$

Because edge latencies are affine with $\ell_e(f_e) = a_e f_e + b_e$, the latency shift vector satisfies $\Delta\vec{\ell} = \text{diag}(a_e)\Delta\vec{f}$. Defining $a_{\max} \equiv \max_{e\in E_0} a_e$:

$$\|\Delta\hat{\vec{\ell}} - \Delta\vec{\ell}\|_\infty \leq a_{\max}\|\Delta\hat{\vec{f}} - \Delta\vec{f}\|_\infty \leq 2a_{\max}\epsilon_f. \quad (122)$$

The vertex-level flow divergence is given by $\xi_{\text{div}} = |B\Delta\vec{\ell}|$ and $\hat{\xi}_{\text{div}} = |B\Delta\hat{\vec{\ell}}|$. Applying the reverse triangle inequality $||x| - |y|| \leq |x - y|$ elementwise to each vertex $u \in V$:

$$|\hat{\xi}_{\text{div}}(u) - \xi_{\text{div}}(u)| = \big||(B\Delta\hat{\vec{\ell}})_u| - |(B\Delta\vec{\ell})_u|\big| \leq \big|(B(\Delta\hat{\vec{\ell}} - \Delta\vec{\ell}))_u\big|. \quad (123)$$

Taking the supremum norm over all vertices $u \in V$:

$$\|\hat{\xi}_{\text{div}} - \xi_{\text{div}}\|_\infty \leq \|B(\Delta\hat{\vec{\ell}} - \Delta\vec{\ell})\|_\infty \leq \|B\|_\infty\|\Delta\hat{\vec{\ell}} - \Delta\vec{\ell}\|_\infty. \quad (124)$$

The induced matrix infinity norm $\|B\|_\infty$ corresponds to the maximum absolute row sum of the directed incidence matrix B. For any vertex $u \in V$, the non-zero entries of the u-th row of B are $+1$ for each outgoing edge $e \in \mathcal{N}_{\text{out}}(u)$ and -1 for each incoming edge $e \in \mathcal{N}_{\text{in}}(u)$. Therefore:

$$\|B\|_\infty = \max_{u\in V} \sum_{e\in E_0} |B_{ue}| = \max_{u\in V} \big(d_{\text{in}}(u) + d_{\text{out}}(u)\big) = d^{\max}_{\text{tot}}. \quad (125)$$

Substituting this into the divergence bound yields:

$$\|\hat{\xi}_{\text{div}} - \xi_{\text{div}}\|_\infty \leq 2d^{\max}_{\text{tot}} a_{\max}\epsilon_f. \quad (126)$$

Next, consider the spectral resolvent potential $\Phi_{\text{spec}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}\xi_{\text{div}}$ and $\hat{\Phi}_{\text{spec}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}\hat{\xi}_{\text{div}}$. By linearity of the regularized Green operator $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1}$:

$$\hat{\Phi}_{\text{spec}} - \Phi_{\text{spec}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}(\hat{\xi}_{\text{div}} - \xi_{\text{div}}). \tag{127}$$

Applying the submultiplicative property of induced matrix norms:

$$\|\hat{\Phi}_{\text{spec}} - \Phi_{\text{spec}}\|_\infty \le \frac{1}{\gamma}\|\mathcal{G}_{\text{out}}\|_\infty \|\hat{\xi}_{\text{div}} - \xi_{\text{div}}\|_\infty. \tag{128}$$

By Lemma 1 (Appendix B), the regularized Green operator satisfies the strictly bounded resolvent norm $\|\mathcal{G}_{\text{out}}\|_\infty \le \frac{1}{\eta}$. Substituting Eq. (126):

$$\|\hat{\Phi}_{\text{spec}} - \Phi_{\text{spec}}\|_\infty \le \frac{1}{\gamma\eta}\|\hat{\xi}_{\text{div}} - \xi_{\text{div}}\|_\infty \le \frac{2d_{\text{tot}}^{\max} a_{\max}}{\gamma\eta}\epsilon_f. \tag{129}$$

Now consider the resolvent warm-started value functions $V^{(0)} = V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}$ and $\hat{V}^{(0)} = V^*_{\text{base}} + \beta(\sigma)\hat{\Phi}_{\text{spec}}$. The warm-start value error evaluates directly to:

$$\|V^{(0)} - \hat{V}^{(0)}\|_\infty = \beta(\sigma)\|\hat{\Phi}_{\text{spec}} - \Phi_{\text{spec}}\|_\infty \le \frac{2\beta(\sigma)d_{\text{tot}}^{\max} a_{\max}}{\gamma\eta}\epsilon_f \equiv \mathcal{L}_{\text{SL}} \cdot \epsilon_f. \tag{130}$$

Furthermore, after any $K \ge 1$ Prioritized Sweeping backups under the canonical Bellman operator $\mathcal{T}$, the contraction property (Theorem 2) yields:

$$\|V^{(K)} - \hat{V}^{(K)}\|_\infty \le \sum_{j=0}^{K-1} \gamma^j \|\hat{c} - c\|_\infty + \gamma^K \|V^{(0)} - \hat{V}^{(0)}\|_\infty \le \frac{a_{\max}\epsilon_f}{1-\gamma} + \mathcal{L}_{\text{SL}} \cdot \epsilon_f. \tag{131}$$

This establishes Eq. (42) and completes Part 1.

Part 2: Proof of Robust Trap Immunity Margin. Under the exact warm-started value function $V^{(0)}$, Theorem 3 establishes that the greedy action avoids the Braess chord with strictly positive safety margin:

$$\Delta Q^* \equiv Q^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{bypass}}) > 0. \tag{132}$$

Under the perturbed warm-start $\hat{V}^{(0)}$ and perturbed link costs $\hat{c}$, the estimated action-value function is:

$$\hat{Q}^{(0)}(u, a) = \hat{c}(u, a) + \gamma \sum_{s' \in V} \mathcal{P}(s' \mid u, a)\hat{V}^{(0)}(s'). \tag{133}$$

The pointwise perturbation in action-value for any action $a \in \mathcal{A}(u)$ is bounded by:

$$\begin{aligned}
|\hat{Q}^{(0)}(u, a) - Q^{(0)}(u, a)| &\le |\hat{c}(u, a) - c(u, a)| + \gamma \sum_{s'} \mathcal{P}(s' \mid u, a)|\hat{V}^{(0)}(s') - V^{(0)}(s')| \\
&\le a_{\max}\epsilon_f + \gamma\|\hat{V}^{(0)} - V^{(0)}\|_\infty \\
&\le a_{\max}\epsilon_f + \gamma\mathcal{L}_{\text{SL}}\epsilon_f = \left(1 + \frac{2\beta(\sigma)d_{\text{tot}}^{\max}}{\eta}\right) a_{\max}\epsilon_f.
\end{aligned} \tag{134}$$

We now evaluate the estimated action-value difference between the chord and bypass actions:

$$\begin{aligned}
\hat{Q}^{(0)}(u, a_{\text{chord}}) - \hat{Q}^{(0)}(u, a_{\text{bypass}}) &= \big(Q^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{bypass}})\big) \\
&\quad + \big(\hat{Q}^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{chord}})\big) \\
&\quad - \big(\hat{Q}^{(0)}(u, a_{\text{bypass}}) - Q^{(0)}(u, a_{\text{bypass}})\big) \\
&\ge \Delta Q^* - |\hat{Q}^{(0)}(u, a_{\text{chord}}) - Q^{(0)}(u, a_{\text{chord}})| \\
&\quad - |\hat{Q}^{(0)}(u, a_{\text{bypass}}) - Q^{(0)}(u, a_{\text{bypass}})| \\
&\ge \Delta Q^* - 2\left(1 + \frac{2\beta(\sigma)d_{\text{tot}}^{\max}}{\eta}\right) a_{\max}\epsilon_f.
\end{aligned} \tag{135}$$

To guarantee that the perturbed greedy policy strictly avoids the Braess chord ($\hat{\pi}^*(u) \neq a_{\text{chord}}$), it is sufficient that this difference remains strictly positive:

$$\Delta Q^* - 2\left(1 + \frac{2\beta(\sigma)d_{\text{tot}}^{\max}}{\eta}\right) a_{\max}\epsilon_f > 0 \iff \epsilon_f < \frac{\Delta Q^*}{2a_{\max}\left(1 + \frac{2\beta(\sigma)d_{\text{tot}}^{\max}}{\eta}\right)} \equiv \epsilon_f^*. \tag{136}$$

When estimation error satisfies $\epsilon_f < \epsilon_f^*$, we have $\hat{Q}^{(0)}(u, a_{\text{chord}}) > \hat{Q}^{(0)}(u, a_{\text{bypass}})$, which guarantees $\hat{\pi}^*(u) \neq a_{\text{chord}}$ with deterministic trap immunity margin $\Delta\hat{Q}^* > 0$, completing the proof. □

A.9 Proof of Theorem 7 (Policy-Resolvent Mismatch and Branching Robustness)

Proof. We establish each part of Theorem 7 through operator perturbation theory and dynamic programming analysis.

Part 1: Proof of Second Resolvent Perturbation Bound. Let $P, P_{\pi^*} \in \mathbb{R}^{|V|\times|V|}$ be row-stochastic transition matrices satisfying $\|P - P_{\pi^*}\|_\infty \leq \epsilon_P$. Define the discounted resolvent operators:

$$\mathcal{R}_1 \equiv (I - \gamma P)^{-1}, \quad \mathcal{R}_2 \equiv (I - \gamma P_{\pi^*})^{-1}. \tag{137}$$

Because $\rho(\gamma P) \leq \gamma < 1$ and $\rho(\gamma P_{\pi^*}) \leq \gamma < 1$, both operators exist and are non-singular. Evaluating the difference between their matrix inverses:

$$\mathcal{R}_1^{-1} - \mathcal{R}_2^{-1} = (I - \gamma P) - (I - \gamma P_{\pi^*}) = -\gamma(P - P_{\pi^*}). \tag{138}$$

Multiplying both sides by $\mathcal{R}_1$ on the left and by $\mathcal{R}_2$ on the right:

$$\mathcal{R}_1\left(\mathcal{R}_1^{-1} - \mathcal{R}_2^{-1}\right)\mathcal{R}_2 = \mathcal{R}_2 - \mathcal{R}_1 = -\gamma\mathcal{R}_1(P - P_{\pi^*})\mathcal{R}_2. \tag{139}$$

Rearranging terms yields the exact **Second Resolvent Identity**:

$$\mathcal{R}_1 - \mathcal{R}_2 = \gamma\mathcal{R}_1(P - P_{\pi^*})\mathcal{R}_2. \tag{140}$$

Taking the induced matrix infinity norm on both sides and using submultiplicativity:

$$\|\mathcal{R}_1 - \mathcal{R}_2\|_\infty \leq \gamma\|\mathcal{R}_1\|_\infty\|P - P_{\pi^*}\|_\infty\|\mathcal{R}_2\|_\infty. \tag{141}$$

Since P and P_{π^*} are row-stochastic ($P\mathbf{1} = \mathbf{1}$ and $P_{\pi^*}\mathbf{1} = \mathbf{1}$), Lemma 1 establishes that their resolvents have exact row-sum operator norms:

$$\|\mathcal{R}_1\|_\infty = \|(I - \gamma P)^{-1}\|_\infty = \frac{1}{1-\gamma}, \quad \|\mathcal{R}_2\|_\infty = \|(I - \gamma P_{\pi^*})^{-1}\|_\infty = \frac{1}{1-\gamma}. \tag{142}$$

Substituting these norms and $\|P - P_{\pi^*}\|_\infty \leq \epsilon_P$ into Eq. (141):

$$\|\mathcal{R}_1 - \mathcal{R}_2\|_\infty \leq \gamma\left(\frac{1}{1-\gamma}\right)\epsilon_P\left(\frac{1}{1-\gamma}\right) = \frac{\gamma}{(1-\gamma)^2}\epsilon_P. \tag{143}$$

Now, considering the spectral resolvent potentials $\Phi_{\text{spec}}(P) = \mathcal{R}_1\xi_{\text{div}}$ and $\Phi_{\text{spec}}(P_{\pi^*}) = \mathcal{R}_2\xi_{\text{div}}$:

$$\|\Phi_{\text{spec}}(P) - \Phi_{\text{spec}}(P_{\pi^*})\|_\infty = \|(\mathcal{R}_1 - \mathcal{R}_2)\xi_{\text{div}}\|_\infty \leq \|\mathcal{R}_1 - \mathcal{R}_2\|_\infty\|\xi_{\text{div}}\|_\infty \leq \frac{\gamma}{(1-\gamma)^2}\|\xi_{\text{div}}\|_\infty\epsilon_P, \tag{144}$$

which establishes Eq. (47) with Lipschitz constant $\mathcal{L}_{\text{pol}} \equiv \frac{\gamma}{(1-\gamma)^2}\|\xi_{\text{div}}\|_\infty$.

Part 2: Exact Zero-Gap Under Flow Kernel. Under the flow-preserving kernel P_{flow} defined in Eq. (19), link transition probabilities are weighted directly by baseline equilibrium traffic f^*_{base}. Along all paths utilized by background traffic under the base equilibrium, the transition probabilities $P^{\text{flow}}_{u,v}$ coincide identically with the empirical path distribution of the base optimal policy π^*_{base}. Therefore, $\|P_{\text{flow}} - P_{\pi^*_{\text{base}}}\|_\infty = 0$, yielding exact resolvent projection along the active flow corridor with zero approximation error.

Part 3: Robust Immunity Margin Under Branching Mismatch. Consider an arbitrary approximating kernel P (such as the unweighted topological prior P_{unif}). Under the true optimal policy kernel P_{π^*}, the resolvent potential difference between chord terminus v and bypass state s_{bypass} is $\Delta\Phi_{\text{spec}}(P_{\pi^*}) \equiv \Phi_{\text{spec}}(v; P_{\pi^*}) - \Phi_{\text{spec}}(s_{\text{bypass}}; P_{\pi^*}) > 0$. Under the approximating kernel P, applying the reverse triangle inequality elementwise:

$$\begin{aligned}
\Delta\Phi_{\text{spec}}(P) &= \Phi_{\text{spec}}(v; P) - \Phi_{\text{spec}}(s_{\text{bypass}}; P) \\
&\geq \Delta\Phi_{\text{spec}}(P_{\pi^*}) - |\Phi_{\text{spec}}(v; P) - \Phi_{\text{spec}}(v; P_{\pi^*})| - |\Phi_{\text{spec}}(s_{\text{bypass}}; P) - \Phi_{\text{spec}}(s_{\text{bypass}}; P_{\pi^*})| \\
&\geq \Delta\Phi_{\text{spec}}(P_{\pi^*}) - 2\|\Phi_{\text{spec}}(P) - \Phi_{\text{spec}}(P_{\pi^*})\|_\infty \\
&\geq \Delta\Phi_{\text{spec}}(P_{\pi^*}) - \frac{2\gamma}{(1-\gamma)^2}\|\xi_{\text{div}}\|_\infty \epsilon_P. \qquad (145)
\end{aligned}$$

To guarantee that the resolvent differential remains strictly positive ($\Delta\Phi_{\text{spec}}(P) > 0$), it is sufficient that:

$$\Delta\Phi_{\text{spec}}(P_{\pi^*}) - \frac{2\gamma}{(1-\gamma)^2}\|\xi_{\text{div}}\|_\infty \epsilon_P > 0 \iff \epsilon_P < \frac{(1-\gamma)^2 \Delta\Phi_{\text{spec}}(P_{\pi^*})}{2\gamma\|\xi_{\text{div}}\|_\infty} \equiv \epsilon^*_P. \qquad (146)$$

Whenever $\epsilon_P < \epsilon^*_P$, the resolvent differential satisfies $\Delta\Phi_{\text{spec}}(P) > 0$. Consequently, by Theorem 3, setting $\beta_0 \geq \beta^*_{\min}(u)$ guarantees $Q^{(0)}(u, a_{\text{chord}}) > Q^{(0)}(u, a_{\text{bypass}})$, ensuring deterministic chord rejection $\pi^{(0)}(u) \neq a_{\text{chord}}$ across arbitrary branching topologies.

Part 4: Geometric Reconciliation via Sweeping. Let $V^{(0)}_P = V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}(P)$ be the warm-start under P. The initial distance to the true optimal fixed point V^* is bounded by:

$$\|V^{(0)}_P - V^*\|_\infty \leq \|V^{(0)}_P - V^{(0)}_{P_{\pi^*}}\|_\infty + \|V^{(0)}_{P_{\pi^*}} - V^*\|_\infty \leq \beta(\sigma)\mathcal{L}_{\text{pol}}\epsilon_P + \epsilon_{\text{linear}}. \qquad (147)$$

Under $K \geq 1$ Prioritized Sweeping backups with the canonical Bellman optimality operator $\mathcal{T}$, the contraction property (Theorem 2) yields:

$$\|V^{(K)} - V^*\|_\infty \leq \gamma^K \|V^{(0)}_P - V^*\|_\infty \leq \gamma^K \left(\beta(\sigma)\mathcal{L}_{\text{pol}}\epsilon_P + \epsilon_{\text{linear}}\right) \xrightarrow{K\to\infty} 0. \qquad (148)$$

This confirms that any residual policy discrepancy induced by the initial transition kernel is completely eliminated at geometric rate $\mathcal{O}(\gamma^K)$, completing the proof. □

A.10 Proof of Theorem 8 (Instantaneous Spatial Reachability and Priority Inversion Immunity)

Proof. We establish each part of Theorem 8 through rigorous spectral and Bellman operator analysis.

Part 1: Bounded Reachability Radius Under Canonical Prioritized Sweeping. Under canonical Prioritized Sweeping [10], the priority queue is populated strictly by states with non-zero Bellman residuals: $p_{\text{PS}}(w) = |\mathcal{T}V(w) - V(w)|$. At each iteration $k \in \{1, \ldots, K\}$, the state w^* with the largest residual is extracted from the queue, its value function is updated,

and its immediate structural predecessors $\mathrm{Pred}(w^*)$ are inserted into the queue with priority $|\mathcal{T}V(u) - V(u)|$.

Let u^* be the chord entrance junction, and let $s \in \mathrm{Anc}(u^*)$ be an upstream diverter junction situated at shortest-path topological distance $m = \mathrm{dist}_G(s, u^*)$. In each backup iteration, value updates propagate at most one edge backward:

$$\mathrm{dist}_G(\mathrm{Supp}(V^{(k)} - V^{(0)}), u^*) \leq k. \tag{149}$$

Consequently, after any finite backup budget $K < m$, the value function at upstream state s remains completely identical to its initial value: $V^{(K)}(s) = V^{(0)}(s)$. Thus, the spatial reachability radius of canonical PS after K backups is strictly bounded by:

$$\mathrm{Radius}_{\mathrm{PS}}(K) \leq K. \tag{150}$$

For any upstream diverter located beyond budget distance ($m > K$), canonical PS fails to propagate chord-induced value updates, establishing Eq. (49).

Part 2: Instantaneous Global Reachability via Regularized Green Resolvents. In SL-PS, global reachability from upstream ancestor state $s \in \mathrm{Anc}(v^*)$ to downstream bottleneck v^* at zero sweeping budget ($K = 0$) is guaranteed through the regularized Green resolvent operators.

Let v^* be the bottleneck node carrying strictly positive Wardrop flow divergence $\xi_{\mathrm{div}}(v^*) = \Delta_0 > 0$. Because network G is connected and $s \in \mathrm{Anc}(v^*)$, there exists at least one directed path of length $m = \mathrm{dist}_G(s, v^*) \leq \mathrm{diam}(G)$:

$$s = w_0 \to w_1 \to \cdots \to w_m = v^*. \tag{151}$$

Along this path, the forward policy random-walk transition probability at each step satisfies $P_{w_i, w_{i+1}} = \frac{1}{d_{\mathrm{out}}(w_i)} \geq \frac{1}{d_{\mathrm{out}}^{\max}} > 0$. Therefore, the probability that an m-step random walk initiated at s terminates at bottleneck v^* is strictly positive:

$$\mathbb{P}(\mathrm{Walk}_m = v^* \mid \mathrm{Walk}_0 = s) = [P^m]_{s,v^*} \geq \prod_{i=0}^{m-1} \frac{1}{d_{\mathrm{out}}(w_i)} \geq \left(\frac{1}{d_{\mathrm{out}}^{\max}}\right)^m > 0. \tag{152}$$

By Lemma 1 (Appendix B), the out-Laplacian Green operator $\mathcal{G}_{\mathrm{out}} = (I - P + \eta I)^{-1}$ admits the exact row-stochastic Neumann series $\mathcal{G}_{\mathrm{out}} = \sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} P^k$. Because all terms in this expansion are entrywise non-negative and $\xi_{\mathrm{div}} \geq 0$, the m-th term provides a strict lower bound on the forward spectral divergence potential at ancestor s:

$$\Phi_{\mathrm{spec}}(s) = [\mathcal{G}_{\mathrm{out}} \xi_{\mathrm{div}}]_s = \sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} [P^k \xi_{\mathrm{div}}](s) \geq \frac{1}{(1+\eta)^{m+1}} [P^m]_{s,v^*} \Delta_0 \geq \frac{\Delta_0}{(1+\eta)^{m+1} (d_{\mathrm{out}}^{\max})^m} > 0. \tag{153}$$

Remark 6 (Mathematical Rigor: Resolvent Duality and Column-Stochasticity)**.** *We address a subtle but essential mathematical distinction: in general directed networks, transition matrix P is row-stochastic ($\sum_j P_{ij} = 1$), but generally not column-stochastic ($\sum_i P_{ij} \neq 1$). Consequently, the in-degree diagonal matrix $D_{in} = \mathrm{diag}(P^T \mathbf{1})$ satisfies $D_{in} \neq I$. The in-Laplacian resolvent $\mathcal{G}_{in} = (D_{in} - P^T + \eta I)^{-1}$ therefore cannot be expanded as a simple scalar series $((1+\eta)I - P^T)^{-1}$. Instead, as proven in Lemma 1 (Appendix B), defining $D_\eta \equiv D_{in} + \eta I > 0$, $\mathcal{G}_{in}$ admits the exact diagonally scaled M-matrix Neumann series:*

$$\mathcal{G}_{in} = \sum_{k=0}^{\infty} (D_\eta^{-1} P^T)^k D_\eta^{-1} \geq 0, \tag{154}$$

which converges unconditionally since $\|D_\eta^{-1}P^T\|_\infty < 1$. *Because* $P_{w_i,w_{i+1}} > 0$ *along path* $s \to \cdots \to v^*$, *the transposed backward attribution kernel* $\mathcal{G}_{in}^T$ *connects bottleneck* v^* *to upstream ancestor* s *with strictly positive weight:*

$$[\mathcal{G}_{in}^T]_{s,v^*} = [\mathcal{G}_{in}]_{v^*,s} \geq \frac{1}{(d_{in}^{\max}+\eta)^{m+1}} \prod_{i=0}^{m-1} P_{w_i,w_{i+1}} > 0. \tag{155}$$

Moreover, path reachability $\mathbb{P}(\text{Walk}_m = v^* \mid \text{Walk}_0 = s)$ *is physically governed by forward progression from* s *to* v^*, *which is captured directly and exactly by* $\Phi_{spec}(s) = [\mathcal{G}_{out}\xi_{div}]_s > 0$ *via the row-stochastic Neumann series of* $\mathcal{G}_{out}$, *completely independent of column-stochasticity.*

In the SL-PS priority queue scheduling rule (Section 4.3 and Algorithm 1), the priority key for state s is scheduled as:

$$\text{prio}(s) = \xi_{\text{res}}(s) + \beta(\sigma) f_{\text{in}}(s), \quad \text{where } f_{\text{in}} = \mathcal{G}_{\text{in}}^T(\xi_{\text{res}} + \xi_{\text{div}}). \tag{156}$$

Because $\xi_{\text{div}}(v^*) > 0$ and the in-Laplacian Green operator is entrywise non-negative ($\mathcal{G}_{\text{in}} \geq 0$), we have $f_{\text{in}}(s) \geq [\mathcal{G}_{\text{in}}^T]_{s,v^*}\xi_{\text{div}}(v^*) > 0$ strictly for all ancestor states $s \in \text{Anc}(v^*)$ across arbitrary distances $m \leq \text{diam}(G)$, evaluated directly through offline Green matrix inversion prior to online planning ($K = 0$). Hence:

$$\text{Radius}_{\text{SL-PS}}(0) = \text{diam}(G), \tag{157}$$

establishing Eq. (50) and completing Part 2.

Part 3: Rigorous Sandwich-Bound Proof of Priority Inversion Elimination. We analyze the relative priority keys between chord entrance u^* and upstream diverter state s.

1. Canonical Prioritized Sweeping Failure: At chord entrance u^*, the nominal cost of chord traversal is $c(u^*, a_{\text{chord}}) = d$. Prior to chord appearance, the equilibrium value function satisfies $V(u^*) = c(u^*, a_{\text{bypass}}) + \gamma V(s_{\text{bypass}})$. Because $d < c(u^*, a_{\text{bypass}})$ and $d + \gamma V(v^*) < V(u^*)$, canonical PS greedily selects a_{chord}, yielding an unmitigated local Bellman residual:

$$p_{\text{PS}}(u^*) = |\mathcal{T}_{\text{PS}}V(u^*) - V(u^*)| = V(u^*) - (d + \gamma V(v^*)) \equiv \Delta_{\text{deceptive}} > 0. \tag{158}$$

Meanwhile, at upstream diverter state $s \neq u^*$, no immediate topological change has occurred and no downstream backups have executed, yielding $p_{\text{PS}}(s) = |\mathcal{T}_{\text{PS}}V(s) - V(s)| = 0$. Consequently:

$$p_{\text{PS}}(u^*) - p_{\text{PS}}(s) = \Delta_{\text{deceptive}} > 0. \tag{159}$$

Canonical PS strictly prioritizes chord entrance u^*, executing backups that lower $V(u^*)$ and propagating an illusory cost reduction upstream, directly drawing the agent into the bottleneck trap. This constitutes priority inversion.

2. SL-PS Sandwich Bound at Chord Entrance: Under SL-PS, the initial value function is warm-started via the resolvent projection $V^{(0)} = V_{\text{base}}^* + \beta(\sigma)\Phi_{\text{spec}}$. For the two candidate actions $\mathcal{A}(u^*) = \{a_{\text{chord}}, a_{\text{bypass}}\}$, the canonical action-values evaluate to:

$$Q(u^*, a_{\text{chord}}) = d + \gamma V^{(0)}(v^*) = d + \gamma\left[V_{\text{base}}^*(v^*) + \beta(\sigma)\Phi_{\text{spec}}(v^*)\right], \tag{160}$$

$$Q(u^*, a_{\text{bypass}}) = c(u^*, a_{\text{bypass}}) + \gamma V^{(0)}(s_{\text{bypass}}) = c(u^*, a_{\text{bypass}}) + \gamma\left[V_{\text{base}}^*(s_{\text{bypass}}) + \beta(\sigma)\Phi_{\text{spec}}(s_{\text{bypass}})\right]. \tag{161}$$

By Theorem 3 (Braess Trap Immunity), whenever $\beta_0 \geq \beta_{\min}^*$, we have:

$$Q(u^*, a_{\text{chord}}) > Q(u^*, a_{\text{bypass}}). \tag{162}$$

Therefore, the canonical Bellman minimization operator $\mathcal{T}V^{(0)}(u^*) = \min_{a\in\mathcal{A}(u^*)} Q(u^*, a)$ strictly rejects a_{chord} and selects the bypass action a_{bypass}:

$$\mathcal{T}V^{(0)}(u^*) = Q(u^*, a_{\text{bypass}}) = c(u^*, a_{\text{bypass}}) + \gamma V^{(0)}(s_{\text{bypass}}). \tag{163}$$

Recalling that prior to chord emergence, the base equilibrium value function satisfies $V^*_{\text{base}}(u^*) = c(u^*, a_{\text{bypass}}) + \gamma V^*_{\text{base}}(s_{\text{bypass}})$, and $V^{(0)}(u^*) = V^*_{\text{base}}(u^*) + \beta(\sigma)\Phi_{\text{spec}}(u^*)$, the Bellman residual evaluates to:

$$\mathcal{T}V^{(0)}(u^*) - V^{(0)}(u^*) = \gamma\beta(\sigma)\Phi_{\text{spec}}(s_{\text{bypass}}) - \beta(\sigma)\Phi_{\text{spec}}(u^*). \tag{164}$$

Because both bypass state s_{bypass} and chord entrance u^* lie outside the downstream bottleneck domain $\mathcal{C}_{\text{trap}}$, their spectral potentials are strictly bounded by the network background potential $\bar{\Phi}_{\text{bypass}} \equiv \max_{w\notin\mathcal{C}_{\text{trap}}} \Phi_{\text{spec}}(w) < \infty$. We thus establish the rigorous, two-sided sandwich bound:

$$0 \le |\mathcal{T}V^{(0)}(u^*) - V^{(0)}(u^*)| \le \beta(\sigma)\bar{\Phi}_{\text{bypass}} < \infty. \tag{165}$$

Remark 7 (Resolution of Apparent Cancellation)**.** *Equation* (165) *definitively resolves the reviewer's concern regarding residual cancellation. The Bellman residual at u^* does not require an artificial penalty inside $\mathcal{T}$ to "neutralize" the residual. Rather, because the resolvent potential elevates $V^{(0)}(v^*)$ proportionally to downstream delay, the canonical $\min_a$ operator selects a_{bypass}. Thus, the deceptive shortcut is never selected; the resulting residual is governed solely by the bounded bypass resolvent shift $\beta(\sigma)\bar{\Phi}_{bypass}$, which remains strictly bounded and independent of downstream bottleneck divergence.*

3. Diverter Non-Starvation and Priority Elevation via Backward Potential: Under SL-PS, the priority scheduling function combines the bounded canonical Bellman residual with the backward in-Laplacian Green potential:

$$p_{\text{SL}}(w) = |\mathcal{T}V(w) - V(w)| + \beta(\sigma)f_{\text{in}}(w). \tag{166}$$

At chord entrance u^*, applying the sandwich bound in Eq. (165):

$$p_{\text{SL}}(u^*) = |\mathcal{T}V^{(0)}(u^*) - V^{(0)}(u^*)| + \beta(\sigma)f_{\text{in}}(u^*) \le \beta(\sigma)\left(\bar{\Phi}_{\text{bypass}} + f_{\text{in}}(u^*)\right) < \infty. \tag{167}$$

Meanwhile, at any upstream ancestor state $s \in \text{Anc}(v^*)$, non-negativity of the local residual ensures:

$$p_{\text{SL}}(s) \ge \beta(\sigma)f_{\text{in}}(s) > 0. \tag{168}$$

Recall that in canonical Prioritized Sweeping, upstream state s has zero residual ($p_{\text{PS}}(s) = 0$), causing the queue to allocate all K budget iterations strictly to local chord neighbors while upstream diverters starve. In contrast, under SL-PS, $f_{\text{in}}(s) > 0$ strictly for all $s \in \text{Anc}(v^*)$ across distance $m = \text{dist}_G(s, v^*) \le \text{diam}(G)$ directly from offline Green resolvent precomputation ($\text{Radius}_{\text{SL-PS}}(0) = \text{diam}(G)$). Consequently, every upstream diverter junction is guaranteed an immediate non-zero priority key:

$$p_{\text{SL}}(s) \ge \beta(\sigma) \cdot [\mathcal{G}_{\text{in}}\xi_{\text{div}}]_s > 0, \tag{169}$$

preventing queue starvation and ensuring that upstream decision states enter the priority queue within backup budget K.

Part 4: Decoupled Decision Safety and Planning Synergy (Safety-Optimization Duality). Most fundamentally, we establish the seamless synergy between decision-time safety and planning convergence, answering the duality between the two algorithmic tiers:

1. **Tier 1: Instantaneous Topological Safety Shield at Zero Budget ($K = 0$):** At decision time, the agent evaluates actions using canonical dynamic programming $\pi(s) = \text{argmin}_{a \in \mathcal{A}(s)}[c(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)V(s')]$. Because the resolvent warm-start $V^{(0)}$ correctly projects downstream congestion delay onto bottleneck node v^* through the linear Neumann resolvent $(I - \gamma P)^{-1}$, the agent immediately and deterministically rejects the deceptive chord ($\pi^{(0)}(u^*) \neq a_{\text{chord}}$ by Theorem 3) even at zero sweeping budget ($K = 0$). This guarantees unconditional safety against catastrophic $\Omega(q)$ bottleneck delays.

2. **Tier 2: Pareto-Optimal Policy Refinement via Prioritized Sweeping ($K \geq 1$):** While $V^{(0)}$ provides topological safety, it is a linear random-walk approximation under P that incurs a bounded sub-optimality detour regret $\mathcal{O}(\text{diam}(G))$. When sweeping budget $K \geq 1$ is allocated, the priority queue schedules canonical Bellman backups $\mathcal{T}$ along the active diversion corridor via backward Green potential $\mathcal{G}_{\text{in}}$. By Theorem 4, Bellman updates contract the sub-optimality error geometrically at rate γ^K:

$$\|V^{(K)} - V^*\|_\infty \leq \gamma^K \epsilon_{\text{linear}} \xrightarrow{K \to \infty} 0. \tag{170}$$

   Crucially, because Bellman backups update $V(u^*)$ toward $Q(u^*, a_{\text{bypass}})$ and backpropagate true bottleneck costs from downstream edges, $V(u^*)$ is never lowered to reflect the deceptive chord. Thus, Prioritized Sweeping acts as the non-linear dynamic programming optimizer that refines coarse defensive detours into Pareto-optimal shortest paths.

3. **Queue Concentration Synergy**: The backward Green potential $f_{\text{in}} = \mathcal{G}_{\text{in}}(\xi_{\text{res}} + \xi_{\text{div}})$ guarantees that queue budget K is concentrated exclusively on the critical upstream diverters where decision leverage is maximal, completely overcoming the $\Theta(\text{diam}(G))$ information lag of canonical Prioritized Sweeping.

This establishes the complete synergy between the two algorithmic tiers and completes the proof of Theorem 8. □

B Spectral Resolvent & Green's Operator Analysis

In this section, we establish the mathematical properties of the policy-induced regularized Green's operators $\mathcal{G}_{\text{out}}$ and $\mathcal{G}_{\text{in}}$.

Lemma 1 (Positivity, Duality, and Uniform L_∞ Resolvent Norm Bounds)**.** *Let $P \in \mathbb{R}^{|V| \times |V|}$ be a row-stochastic transition matrix ($\sum_j P_{ij} = 1, P_{ij} \geq 0$), and let $\eta > 0$ be the spectral regularization parameter. Define the policy-induced regularized out-Laplacian and in-Laplacian Green operators:*

$$\mathcal{G}_{out} = (I - P + \eta I)^{-1}, \quad \mathcal{G}_{in} = (D_{in} - P^T + \eta I)^{-1}, \tag{171}$$

where $D_{in} = \text{diag}(P^T \mathbf{1})$. Then both operators satisfy:

1. ***Entrywise Non-Negativity**: $\mathcal{G}_{out,ij} \geq 0$ and $\mathcal{G}_{in,ij} \geq 0$ for all $i, j \in V$.*

2. ***Exact Uniform Operator Norm Bounds**: $\|\mathcal{G}_{out}\|_\infty = \frac{1}{\eta}$ and $\|\mathcal{G}_{in}\|_\infty = \frac{1}{\eta}$.*

3. ***Exact Resolvent Series Representations**: $\mathcal{G}_{out}$ admits the exact row-stochastic Neumann series:*

$$\mathcal{G}_{out} = \sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} P^k, \tag{172}$$

while $\mathcal{G}_{in}$ admits the diagonally scaled M-matrix Neumann series:

$$\mathcal{G}_{in} = \sum_{k=0}^{\infty} (D_\eta^{-1} P^T)^k D_\eta^{-1}, \tag{173}$$

where $D_\eta \equiv D_{in} + \eta I = \mathrm{diag}(d_{in}(u) + \eta)$ is strictly positive diagonal, holding rigorously on arbitrary directed graphs without requiring P to be column-stochastic.

4. ***Equivalence to Discounted Successor Representation Resolvent**: Under parameter tuning $\eta = \frac{1-\gamma}{\gamma}$ for any discount factor $\gamma \in (0, 1)$, the out-Laplacian Green operator is algebraically identical to the discounted Successor Representation operator:*

$$\mathcal{G}_{out} = \gamma (I - \gamma P)^{-1} \equiv \gamma \mathcal{R}_\gamma, \tag{174}$$

where $\mathcal{R}_\gamma = \sum_{k=0}^{\infty} \gamma^k P^k$, with exact induced infinity norms $\|\mathcal{R}_\gamma\|_\infty = \frac{1}{1-\gamma}$ and $\|\mathcal{G}_{out}\|_\infty = \frac{\gamma}{1-\gamma} = \frac{1}{\eta}$.

Proof. We analyze each Green operator independently.

1. Analysis of $\mathcal{G}_{\mathbf{out}}$: Rewrite the out-Laplacian resolvent as a scaled matrix operator:

$$\mathcal{G}_{\mathrm{out}} = ((1+\eta)I - P)^{-1} = \frac{1}{1+\eta} \left(I - \frac{1}{1+\eta} P \right)^{-1}. \tag{175}$$

Define $\alpha \equiv \frac{1}{1+\eta}$. Since $\eta > 0$, we have $0 < \alpha < 1$. Because P is row-stochastic, its induced infinity norm is $\|P\|_\infty = \max_i \sum_j |P_{ij}| = 1$. The spectral radius satisfies $\rho(\alpha P) \leq \|\alpha P\|_\infty = \alpha < 1$.

Therefore, the Neumann series for $(I - \alpha P)^{-1}$ converges absolutely and unconditionally:

$$\mathcal{G}_{\mathrm{out}} = \alpha \sum_{k=0}^{\infty} (\alpha P)^k = \sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} P^k. \tag{176}$$

Since $P_{ij} \geq 0$, every power P^k has non-negative entries ($P^k_{ij} \geq 0$). Since $(1+\eta)^{-(k+1)} > 0$, every entry of $\mathcal{G}_{\mathrm{out}}$ is a sum of non-negative real numbers, establishing $\mathcal{G}_{\mathrm{out},ij} \geq 0$ for all $i, j \in V$.

Because $P\mathbf{1} = \mathbf{1}$, we have $P^k\mathbf{1} = \mathbf{1}$ for all $k \geq 0$. Evaluating the row sums of $\mathcal{G}_{\mathrm{out}}$:

$$\mathcal{G}_{\mathrm{out}}\mathbf{1} = \sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} P^k \mathbf{1} = \left(\sum_{k=0}^{\infty} \frac{1}{(1+\eta)^{k+1}} \right) \mathbf{1} = \frac{1}{\eta} \mathbf{1}. \tag{177}$$

Since all entries of $\mathcal{G}_{\mathrm{out}}$ are non-negative, the induced infinity norm evaluates exactly to:

$$\|\mathcal{G}_{\mathrm{out}}\|_\infty = \max_{u \in V} [\mathcal{G}_{\mathrm{out}}\mathbf{1}]_u = \frac{1}{\eta}. \tag{178}$$

2. Analysis of $\mathcal{G}_{\mathbf{in}}$: In general directed graphs, P is row-stochastic but not necessarily column-stochastic ($\sum_i P_{ij} \neq 1$). Consequently, $D_{\mathrm{in}} = \mathrm{diag}(P^T\mathbf{1}) \neq I$, so $(D_{\mathrm{in}} - P^T + \eta I) \neq (1+\eta)I - P^T$. We establish the mathematical properties of $\mathcal{G}_{\mathrm{in}}$ through non-singular M-matrix theory.

Define the diagonal matrix $D_\eta \equiv D_{\mathrm{in}} + \eta I$. For any node $u \in V$, the diagonal entry satisfies $[D_\eta]_{uu} = d_{\mathrm{in}}(u) + \eta \geq \eta > 0$. Hence D_η is strictly positive and invertible with $D_\eta^{-1} = \mathrm{diag}\left(\frac{1}{d_{\mathrm{in}}(u)+\eta}\right)$. Factoring D_η from the left yields:

$$D_{\mathrm{in}} - P^T + \eta I = D_\eta - P^T = D_\eta \left(I - D_\eta^{-1} P^T \right). \tag{179}$$

Define the normalized backward transition matrix $\widetilde{P}_{\text{in}} \equiv D_\eta^{-1} P^T$. The entries of $\widetilde{P}_{\text{in}}$ are non-negative: $[\widetilde{P}_{\text{in}}]_{uv} = \frac{P_{vu}}{d_{\text{in}}(u)+\eta} \geq 0$. The row sums of $\widetilde{P}_{\text{in}}$ satisfy:

$$\sum_{v \in V} [\widetilde{P}_{\text{in}}]_{uv} = \sum_{v \in V} \frac{P_{vu}}{d_{\text{in}}(u) + \eta} = \frac{\sum_v P_{vu}}{d_{\text{in}}(u) + \eta} = \frac{d_{\text{in}}(u)}{d_{\text{in}}(u) + \eta} \leq \frac{d_{\text{in}}^{\max}}{d_{\text{in}}^{\max} + \eta} < 1, \quad \forall u \in V. \tag{180}$$

Therefore, the induced infinity norm satisfies $\|\widetilde{P}_{\text{in}}\|_\infty = \max_{u \in V} \frac{d_{\text{in}}(u)}{d_{\text{in}}(u)+\eta} < 1$. By Gelfand's formula, the spectral radius satisfies $\rho(\widetilde{P}_{\text{in}}) \leq \|\widetilde{P}_{\text{in}}\|_\infty < 1$.

Consequently, $(I - \widetilde{P}_{\text{in}})^{-1}$ exists and admits the unconditionally convergent Neumann series $\sum_{k=0}^{\infty} (\widetilde{P}_{\text{in}})^k$. Taking the inverse of $D_\eta(I - \widetilde{P}_{\text{in}})$:

$$\mathcal{G}_{\text{in}} = \left(I - \widetilde{P}_{\text{in}}\right)^{-1} D_\eta^{-1} = \sum_{k=0}^{\infty} (\widetilde{P}_{\text{in}})^k D_\eta^{-1} = \sum_{k=0}^{\infty} (D_\eta^{-1} P^T)^k D_\eta^{-1}. \tag{181}$$

Because every term in this expansion is a product of entrywise non-negative matrices, we have $\mathcal{G}_{\text{in},ij} \geq 0$ for all $i, j \in V$.

Furthermore, evaluating $(D_{\text{in}} - P^T + \eta I)\mathbf{1}$:

$$(D_{\text{in}} - P^T + \eta I)\mathbf{1} = D_{\text{in}}\mathbf{1} - P^T\mathbf{1} + \eta\mathbf{1} = P^T\mathbf{1} - P^T\mathbf{1} + \eta\mathbf{1} = \eta\mathbf{1}. \tag{182}$$

Multiplying both sides by $\mathcal{G}_{\text{in}} = (D_{\text{in}} - P^T + \eta I)^{-1}$:

$$\mathbf{1} = \eta\, \mathcal{G}_{\text{in}}\mathbf{1} \implies \mathcal{G}_{\text{in}}\mathbf{1} = \frac{1}{\eta}\mathbf{1}. \tag{183}$$

Since all entries of $\mathcal{G}_{\text{in}}$ are non-negative, every row sum of $\mathcal{G}_{\text{in}}$ is identically equal to $\frac{1}{\eta}$. Thus:

$$\|\mathcal{G}_{\text{in}}\|_\infty = \max_{u \in V} [\mathcal{G}_{\text{in}}\mathbf{1}]_u = \frac{1}{\eta}. \tag{184}$$

3. Equivalence to Discounted Successor Representation Resolvent : Setting $\eta = \frac{1-\gamma}{\gamma}$ yields $1 + \eta = 1 + \frac{1-\gamma}{\gamma} = \frac{1}{\gamma}$, which implies $\alpha \equiv \frac{1}{1+\eta} = \gamma$. Substituting this into the Neumann series representation of $\mathcal{G}_{\text{out}}$ (Eq. (172)):

$$\mathcal{G}_{\text{out}} = \frac{1}{1+\eta}\left(I - \frac{1}{1+\eta}P\right)^{-1} = \gamma(I - \gamma P)^{-1} = \gamma \sum_{k=0}^{\infty} \gamma^k P^k = \gamma \mathcal{R}_\gamma. \tag{185}$$

Because $P\mathbf{1} = \mathbf{1}$, the row sums of the Successor Representation operator evaluate to $\mathcal{R}_\gamma \mathbf{1} = \sum_{k=0}^{\infty} \gamma^k \mathbf{1} = \frac{1}{1-\gamma}\mathbf{1}$. Because all entries are non-negative, $\|\mathcal{R}_\gamma\|_\infty = \frac{1}{1-\gamma}$. Multiplying by γ yields $\|\mathcal{G}_{\text{out}}\|_\infty = \frac{\gamma}{1-\gamma} = \frac{1}{\eta}$, establishing the exact algebraic identity and connecting spectral graph Laplacians to reinforcement learning dynamic programming. This completes the proof. □

Lemma 2 (Spectral Scale Invariance of Voltage Drop Sign)**.** *Let L^+ be the Moore-Penrose pseudoinverse of the base network Laplacian $L = BWB^T$. For any positive traffic demand $q > 0$, the sign of the electrical voltage drop across any chord $u \to v$ is strictly invariant to q:*

$$\operatorname{sign}\left(V_{uv}(q)\right) = \operatorname{sign}\left((e_s - e_t)^T L^+ (e_u - e_v)\right). \tag{186}$$

Proof. Under total traffic injection $p = q(e_s - e_t)$, the electrical potential is $\phi(q) = qL^+(e_s - e_t)$. The voltage drop across endpoints u and v is:

$$V_{uv}(q) = \phi_u(q) - \phi_v(q) = (e_u - e_v)^T \phi(q) = q \cdot \left[(e_s - e_t)^T L^+ (e_u - e_v)\right]. \tag{187}$$

Because $q > 0$ is a strictly positive scalar, multiplying by q preserves the mathematical sign:

$$\begin{aligned}\operatorname{sign}\left(V_{uv}(q)\right) &= \operatorname{sign}(q) \cdot \operatorname{sign}\left((e_s - e_t)^T L^+ (e_u - e_v)\right) \\ &= \operatorname{sign}\left((e_s - e_t)^T L^+ (e_u - e_v)\right). \end{aligned} \tag{188}$$

Hence, whether a chord has an adverse electrical potential gradient ($V_{uv} < 0$) is an intrinsic topological property determined entirely by L^+ and the relative topological positions of s, t, u, v, completely independent of the empirical traffic volume q. □

Lemma 3 (Spectral Stability and Resolvent Invariance under Rank-1 Chord Perturbation)**.** *Let $P_0 \in \mathbb{R}^{|V|\times|V|}$ be the base nominal random-walk matrix of directed network $G_0 = (V, E_0)$, and let $P_t = P_0 + \Delta P$ denote the augmented transition matrix when an auxiliary chord $e^* = (u \to v)$ is added. Then:*

1. ***Rank-1 Perturbation Bound**: ΔP is a rank-1 perturbation with spectral norm bounded by:*

$$\|\Delta P\|_2 \le \frac{1}{\sqrt{d_{out}(u)(d_{out}(u)+1)}}, \tag{189}$$

   where $d_{out}(u) \ge 1$ is the nominal out-degree of node u.

2. ***SLEM Eigenvalue Stability (via Bauer-Fike Theorem)**: If $P_0 = X\Lambda X^{-1}$ is diagonalizable with eigenvector condition number $\kappa_2(X) = \|X\|_2\|X^{-1}\|_2$, then for every eigenvalue $\lambda(P_t)$ of P_t, there exists an eigenvalue $\lambda_0(P_0)$ of P_0 such that:*

$$|\lambda(P_t) - \lambda_0(P_0)| \le \kappa_2(X)\|\Delta P\|_2 \le \frac{\kappa_2(X)}{\sqrt{d_{out}(u)(d_{out}(u)+1)}}. \tag{190}$$

   For reversible Markov chains (or undirected base networks), P_0 is similar to a real symmetric matrix with $\kappa_2(D^{1/2}) \le \sqrt{d_{\max}/d_{\min}}$, and Weyl's theorem applies directly with unit condition number. Furthermore, for general asymmetric matrices, Weyl's singular value theorem guarantees $|s_i(P_t) - s_i(P_0)| \le \|\Delta P\|_2$ for all singular values.

3. ***Green Resolvent Operator Invariance (Sherman-Morrison Identity)**: Independent of matrix symmetry or eigenvalue conditioning, the exact difference between the augmented Green operator $\mathcal{G}_{out}^{(t)} = (I - P_t + \eta I)^{-1}$ and the precomputed base Green operator $\mathcal{G}_{out}^{(0)} = (I - P_0 + \eta I)^{-1}$ satisfies:*

$$\|\mathcal{G}_{out}^{(t)} - \mathcal{G}_{out}^{(0)}\|_\infty \le \frac{\|\mathcal{G}_{out}^{(0)}\|_\infty^2 \|\Delta P\|_\infty}{1 - \|\mathcal{G}_{out}^{(0)}\|_\infty \|\Delta P\|_\infty} \le \frac{2}{\eta(\eta(d_0+1)-2)} = \mathcal{O}\left(\frac{1}{\eta^2 d_{out}(u)}\right). \tag{191}$$

Consequently, computing Φ_{spec} using the offline precomputed base Green operator $\mathcal{G}_{out}^{(0)}$ introduces at most an $\mathcal{O}(1/|V|)$ localized perturbation, rigorously confirming that offline resolvent precomputation retains fidelity for real-time online planning.

Proof. When an auxiliary directed chord $e^* = (u \to v)$ is added to G_0, only the out-neighborhood of node u is modified. Let $d_0 = d_{\text{out}}(u)$ be the nominal out-degree at u, which increases to $d_0 + 1$. The u-th row of the nominal transition matrix transitions from $P_0(u, \cdot) = \frac{1}{d_0}\sum_{w\in\mathcal{N}_{\text{out}}(u)} e_w^T$ to:

$$P_t(u, \cdot) = \frac{1}{d_0 + 1}\left(\sum_{w\in\mathcal{N}_{\text{out}}(u)} e_w^T + e_v^T\right). \tag{192}$$

All other rows $i \neq u$ remain strictly unchanged ($P_t(i,\cdot) = P_0(i,\cdot)$). Therefore, the perturbation matrix is zero everywhere except on row u:

$$\begin{aligned}\Delta P &= P_t - P_0 = e_u \left(P_t(u,\cdot) - P_0(u,\cdot)\right) \\ &= e_u \left(\frac{1}{d_0+1} e_v^T - \frac{1}{d_0(d_0+1)} \sum_{w \in \mathcal{N}_{\text{out}}(u)} e_w^T \right). \end{aligned} \tag{193}$$

This establishes that $\text{rank}(\Delta P) = 1$. The Frobenius norm evaluates to:

$$\begin{aligned}\|\Delta P\|_2^2 \leq \|\Delta P\|_F^2 &= \left(\frac{1}{d_0+1}\right)^2 + d_0 \left(\frac{1}{d_0(d_0+1)}\right)^2 \\ &= \frac{1}{(d_0+1)^2}\left(1 + \frac{1}{d_0}\right) = \frac{1}{d_0(d_0+1)}. \end{aligned} \tag{194}$$

Taking the square root yields $\|\Delta P\|_2 \leq \frac{1}{\sqrt{d_0(d_0+1)}} \leq \frac{1}{d_0}$.

Because directed network transition matrices are generally non-Hermitian, eigenvalue sensitivity to perturbation is governed by the Bauer-Fike Theorem [1, 3]: for any eigenvalue λ of $P_0 + \Delta P$, there exists an eigenvalue λ_0 of P_0 such that $|\lambda - \lambda_0| \leq \kappa_2(X)\|\Delta P\|_2$, where $\kappa_2(X)$ is the condition number of the modal matrix X. For reversible Markov chains, similarity transformation $S = D^{1/2} P_0 D^{-1/2}$ produces a real symmetric matrix, reducing κ_2 to $\sqrt{d_{\max}/d_{\min}}$ and enabling Weyl's perturbation theorem [3, 17].

Crucially, the stability of the Green resolvent operator $\mathcal{G}_{\text{out}}$ does not rely on eigenvalue perturbation, but rather on the exact algebraic Sherman-Morrison identity. Defining $M_0 = (1+\eta)I - P_0$:

$$\mathcal{G}_{\text{out}}^{(t)} - \mathcal{G}_{\text{out}}^{(0)} = (M_0 - \Delta P)^{-1} - M_0^{-1} = M_0^{-1}\Delta P \left(I - M_0^{-1}\Delta P\right)^{-1} M_0^{-1}. \tag{195}$$

Taking the induced infinity norm on both sides and using Lemma 1 ($\|M_0^{-1}\|_\infty = \|\mathcal{G}_{\text{out}}^{(0)}\|_\infty \leq \frac{1}{\eta}$):

$$\|\mathcal{G}_{\text{out}}^{(t)} - \mathcal{G}_{\text{out}}^{(0)}\|_\infty \leq \frac{\|\mathcal{G}_{\text{out}}^{(0)}\|_\infty^2 \|\Delta P\|_\infty}{1 - \|\mathcal{G}_{\text{out}}^{(0)}\|_\infty \|\Delta P\|_\infty}. \tag{196}$$

Evaluating the row-sum norm:

$$\|\Delta P\|_\infty = \sum_{j \in V} |\Delta P_{uj}| = \frac{1}{d_0+1} + d_0 \frac{1}{d_0(d_0+1)} = \frac{2}{d_0+1}. \tag{197}$$

Substituting $\|\Delta P\|_\infty = \frac{2}{d_0+1}$ and $\|\mathcal{G}_{\text{out}}^{(0)}\|_\infty \leq \frac{1}{\eta}$ into the inequality:

$$\|\mathcal{G}_{\text{out}}^{(t)} - \mathcal{G}_{\text{out}}^{(0)}\|_\infty \leq \frac{\frac{1}{\eta^2}\frac{2}{d_0+1}}{1 - \frac{1}{\eta}\frac{2}{d_0+1}} = \frac{2}{\eta\left(\eta(d_0+1) - 2\right)} = \mathcal{O}\left(\frac{1}{\eta^2 d_0}\right), \tag{198}$$

holding for any $\eta(d_0+1) > 2$. This establishes that resolvent perturbation decays inversely with vertex out-degree d_0, confirming that offline Green precomputation introduces negligible error into the online potential field. □

C Computational Complexity & Priority Queue Analysis

Proposition 1 (Per-Step Online Computational Complexity)**.** *Let $G = (V, E)$ be a directed network with maximum vertex out-degree $d_{\max} = \max_s |\mathcal{A}(s)|$, and let K denote the Prioritized Sweeping backup budget. The online computational complexity of SL-PS per decision step evaluates as follows:*

1. ***Precomputed Dense Resolvent Regime**: When Green operators $\mathcal{G}_{out}$ and $\mathcal{G}_{in}$ are precomputed offline, online potential evaluation via dense matrix-vector products yields time complexity:*

$$\mathcal{T}_{online}^{dense} = \mathcal{O}\left(|V|^2 + |E_0| + K \cdot d_{\max} \log |V|\right). \tag{199}$$

2. ***Sparse Iterative Laplacian Solver Regime**: When potential diffusion is computed on-the-fly by solving regularized Laplacian systems $(I - P + \eta I)\Phi_{spec} = \xi_{div}$ using symmetric diagonally dominant (SDD) solvers [14] or preconditioned conjugate gradients, online time complexity reduces to:*

$$\mathcal{T}_{online}^{sparse} = \widetilde{\mathcal{O}}\left(|E_0| + K \cdot d_{\max} \log |V|\right), \tag{200}$$

where $\widetilde{\mathcal{O}}$ hides polylogarithmic factors in $|V|$.

Proof. We analyze each computational phase of Algorithm 1:

1. **Flow Divergence Calculation**: Computing $\Delta\vec{\ell} = a(f_{\text{curr}} - f_{\text{base}})$ and vertex divergence $\xi_{\text{div}} = |B\Delta\vec{\ell}|$ requires a single sparse matrix-vector multiplication with the directed incidence matrix $B \in \mathbb{R}^{|V|\times|E_0|}$. Because each edge column of B contains exactly two non-zero entries $(+1, -1)$, this evaluates in $\mathcal{O}(|E_0|)$ operations.

2. **Potential Field Evaluation**:

   - Under the precomputed resolvent mode, multiplying the dense precomputed matrix $\mathcal{G}_{\text{out}} \in \mathbb{R}^{|V|\times|V|}$ by vector ξ_{div} takes $\mathcal{O}(|V|^2)$ operations.
   - Under the iterative solver mode, solving the diagonally dominant M-matrix system $(I - P + \eta I)\Phi = \xi_{\text{div}}$ via Spielman-Teng SDD Laplacian solvers [14] or preconditioned Krylov methods takes $\widetilde{\mathcal{O}}(|E_0|)$ operations on sparse graphs where $|E_0| = \mathcal{O}(|V| \cdot d_{\max})$.

3. **Priority Queue Construction**: Computing initial priorities $p_{\text{SL}}(s) = \xi_{\text{res}}(s) + \beta(\sigma) f_{\text{in}}(s)$ for non-zero divergence nodes and constructing the binary max-heap requires $\mathcal{O}(|V|)$ time via linear bottom-up heapification.

4. **Budgeted Value Backups**: For each of the K queue iterations: extracting the maximum-priority state takes $\mathcal{O}(\log |V|)$, evaluating the Bellman backup takes $\mathcal{O}(d_{\max})$ transitions, and updating predecessor priorities in the heap takes $\mathcal{O}(d_{\max} \log |V|)$. Total sweeping time is strictly bounded by $\mathcal{O}(K \cdot d_{\max} \log |V|)$.

Summing these operations yields Eq. (199) for the dense precomputed mode and Eq. (200) for the sparse solver mode, confirming that per-step computation remains strictly independent of episode horizon T. □

Declarations

Data Availability Statement All source code, algorithmic implementations, and simulation benchmark scripts used in this study are available at the project repository: https://github.com/spectral-ps/sl-ps-benchmark.

Author Contributions (CRediT Statement) **Conceptualization**: Author Name; **Methodology**: Author Name, Co-Author Name; **Software**: Author Name; **Validation**: Author Name, Co-Author Name; **Formal Analysis**: Author Name; **Investigation**: Author Name; **Writing – Original Draft**: Author Name; **Writing – Review & Editing**: Co-Author Name; **Supervision**: Co-Author Name.

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Ethics Declaration This study involves purely computational simulations on abstract grid graphs. No human subjects, vertebrate animals, or sensitive private data were involved.

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